Voiding Function & Dysfunction Rapid Revision

Campbell-Walsh-Wein · Chapters 103, 104, 107, 109, 113, 115

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 3h 13min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 107* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
103	Physiology & Pharmacology of the Bladder and Urethra	51 min
104	Pathophysiology & Classification of LUTD	16 min
107	Urodynamic & Videourodynamic Evaluation	41 min
109	Neuromuscular Dysfunction of the LUT	42 min
113	Pharmacologic Management of Storage & Emptying Failure	19 min
115	Electrical Stimulation & Neuromodulation	18 min
★	Putting it together: voiding dysfunction decisions	6 min
	Total (13454 words, at 70 words/min)	**3h 13min**

103 Physiology and Pharmacology of the Bladder and Urethra

Reading time ≈ 51 min (slow pace)

1 · OVERVIEW

- The lower urinary tract (LUT: bladder + urethra) has two defining functions: **storage** (continence) and **emptying** (micturition).
- Control integrates the central nervous system (CNS), peripheral nerves, detrusor smooth muscle (DSM), urothelium, urethral muscle, pelvic floor and external urethral sphincter (EUS).
- Lower urinary tract dysfunction (LUTD) includes urinary incontinence (UI), lower urinary tract symptoms (LUTS), overactive bladder (OAB), bladder pain syndrome/interstitial cystitis (BPS/IC), neurogenic bladder and underactive bladder (UAB).

> **KEY POINT** **Why single-organ therapy fails:** bladder and urethra act reciprocally, so boosting contractility in detrusor underactivity fails unless the urethra also relaxes. Afferent (sensory) modulation acts on both at once — reduce it for overactivity, augment it for underactivity — which makes it the most promising therapeutic lever.

2 · FUNCTIONAL ANATOMY

Bladder

- **Body** lies above the ureteric orifices; **base** = trigone + bladder neck. The two regions differ in neuromorphology and neuropharmacology but each is internally homogeneous.
- **Male bladder neck**: complete smooth muscle ring with dense sympathetic noradrenergic supply; it closes during ejaculation → **antegrade ejaculation**. Female: no such collar, sparse adrenergic supply.
- **Lamina propria** = proposed "functional centre": unmyelinated nerve plexus with TRPV1 and P2X3 receptors, interstitial cells (myofibroblasts) and a subepithelial capillary plexus (the vessels seen at cystoscopy).
- **Blood flow** falls with distension; intramural tension is the main determinant. In low compliance, pressure rises more and flow falls more. Ischaemia rapidly reduces contractility; ischaemia–reperfusion damages intramural neurons → patchy denervation, as seen in detrusor overactivity (DO).

Stroma, collagen and elastin

- Wall ≈ **50% collagen, 2% elastin** besides muscle. Collagen mostly outside muscle bundles; types I, III, IV predominate.
- **Poorly compliant bladders**: ↑ connective tissue : smooth muscle ratio and ↑ **type III : type I collagen**; collagen fibres lose the ability to change tortuosity → ↓ capacity.

Urethra and sphincters

- Bladder outlet = urethra + pelvic floor. Urethra has smooth muscle (lissosphincter) and an **intramural striated rhabdosphincter (EUS)**, separate from periurethral pelvic floor muscle.
- **Male urethra**: preprostatic (bladder neck) → prostatic → membranous (through pelvic floor, incl. EUS) → bulbous and penile. Male EUS: crescent ventral to prostate proximal to verumontanum → horseshoe distally → crescent at bulbar urethra.
- **Female urethra**: in distal third of anterior vaginal wall; EUS a horseshoe over the ventral urethra; subepithelial vascular plexus forms a **"urethral seal"**.
- Levator ani does not encircle the ventral urethra — its direct continence role is questioned.

Feature	Slow-twitch	Fast-twitch
Role	Sustained resting tone	Rapid extra squeeze on coughing
Fatigue / metabolism	Slow; oxidative	Rapid; anaerobic
Where	Intramural rhabdosphincter (predominant) — gives **~50% of static urethral resistance**	Periurethral pelvic floor (mixed)
Rhabdosphincter mix	Male **65% slow / 35% fast**; female **87% slow / 13% fast**	

- Pelvic floor exercise/electrostimulation may help stress urinary incontinence (SUI) by converting fast to slow fibres (speculative).

3 · UROTHELIUM: BARRIER AND SENSOR

Barrier

- ~7 layers: basal, intermediate, apical **umbrella cells** (largest, hexagonal, sometimes multinucleate).
- **Uroplakins** (unique to umbrella cells) form plaques of **asymmetrical unit membrane (AUM)** — outer leaflet thicker; joined by hinge regions. They provide the barrier and are the binding site of **type 1-fimbriated *E. coli***.
- **Tight junctions** (occludin, claudins, zona occludens-1) block paracellular flow. Stretch lowers transepithelial resistance tenfold (ionic paracellular transport) without changing small-molecule permeability.
- Loss of umbrella cells (cystitis, infective/radiation) → solute leak into lamina propria → symptoms worsened by distension; hyperosmolality excites C-fibre afferents.
- **Glycosaminoglycan (GAG) layer**: controversial. The "antiadherence factor" proved to be a glycoprotein (mucin MUC-1); chondroitin sulfate is the main human GAG. GAG may aid antiadherence but is **not proven** to be the primary impermeability barrier.

Sensor–transducer function

Urothelial cells behave like sensory neurons: they express receptors for bradykinin, neurotrophins (TrkA, p75), purines (P2X, P2Y), noradrenaline (α, β), acetylcholine (ACh; nicotinic, muscarinic), ENaC and transient receptor potential (TRP) channels (TRPV1, TRPV2, TRPV4, TRPM8). On stretch or chemical stimulation they release mediators onto adjacent afferents, directly or via suburothelial myofibroblasts.

Mediator	Released by / when	Effect
ATP	Stretch; ↑ with injury, inflammation, ageing	Excites suburothelial **P2X3** (and P2X2/3) afferents → fullness, pain; P2X3-null mice are hyporeflexic
Nitric oxide (NO)	Especially in inflammation	Little effect on detrusor; inhibits afferent Ca^{2+} channels → net **afferent inhibition**
ACh (non-neuronal)	Basal; ↑ by stretch and ageing	Modulates afferents and DSM; muscarinic receptors on urothelium → antimuscarinics may act partly here
Prostaglandins (PGs)	Urothelium = main source; ↑ in inflammation	Detrusor regulation, cytoprotection
Inhibitory factor (unknown)	Basal	↓ muscarinic contraction; not NO or PG

- Normal afferent thresholds reflect an **ATP (excitatory) : NO (inhibitory) balance**; ATP ↑ in DO and BPS/IC.
- **Suburothelial myofibroblasts**: vimentin+, α-smooth muscle actin+, desmin−; coupled by **connexin 43 (Cx43)** gap junctions; close to C-fibre endings; respond to ATP (P2Y6) and ACh

(M2/M3). In idiopathic DO (IDO), their M2/M3 expression correlates with urgency. NO donors damp them via cyclic guanosine monophosphate (cGMP).

> **KEY POINT** A damaged urothelium produces symptoms twice over — by **solute leak** into the lamina propria and by **altered signalling** (more ATP, less NO) to afferents.

4 · DETRUSOR SMOOTH MUSCLE PHYSIOLOGY

Structure

- Fascicles in random directions coupled as a functional syncytium (unlike the discrete layers of ureter and gut). Cells spindle-shaped, single nucleus, no striations, a few hundred µm × 5–6 µm.
- No sarcomeres: thick (myosin II), thin (actin + tropomyosin + caldesmon) and intermediate (desmin, vimentin) filaments anchor to **dense bodies**.
- Motor supply: postganglionic parasympathetic varicosities release ACh and ATP; not every cell is contacted — **gap junctions** spread excitation.

Property	Skeletal muscle	Smooth muscle
Innervation	Somatic	Autonomic + hormonal, paracrine
Activation	Ca^{2+}–troponin removes tropomyosin inhibition; rapid	Ca^{2+}–calmodulin → **myosin phosphorylation**; slow; "latch state"
Contraction / tone	Phasic; tone neural	Mostly tonic; intrinsic + extrinsic tone

Excitation–contraction coupling

ALGORITHM
ACh → **M3** (Gq) → ↑ $[Ca^{2+}]i$ (L-type Ca^{2+} entry + phospholipase C → IP3 → SR release) → Ca^{2+}–calmodulin → myosin light chain kinase (MLCK) → phosphorylates MLC20 → actin–myosin cross-bridge cycling → contraction → myosin light chain phosphatase (MLCP) dephosphorylates as Ca^{2+} falls → relaxation

- Tone = Ca^{2+} balance + **MLCK : MLCP balance**. **Calcium sensitisation**: RhoA-GTP → Rho kinase (ROK) and protein kinase C (via CPI-17) inhibit MLCP → force rises with no rise in Ca^{2+}. Muscarinic activation recruits ROK.
- **L-type voltage-dependent Ca^{2+} channels (VDCC)** are the main contraction trigger (blocked by nifedipine, diltiazem, verapamil); in mouse DSM nerve-evoked contraction depends on L-type entry alone. T-type: finer control.
- Resting potential −50 to −60 mV. Action potential upstroke = Ca^{2+} inward current; repolarisation = K^+ outward current via SK, **BK** (large-conductance Ca^{2+}-activated K^+) and Kv channels. **BK is the key brake**: Ca^{2+} sparks from ryanodine receptors open BK → hyperpolarisation. BK knockout → OAB phenotype; BK ↓ in neurogenic DO (NDO).

G protein	Receptors	Effect
Gq	M1, M3, M5 ("odd"); α1; angiotensin II; H1; 5-HT2	IP3 + DAG → Ca^{2+} release, PKC → contraction
Gi/o	M2, M4 ("even")	↓ adenylyl cyclase → ↓ cAMP
Gs	β-adrenoceptors (β3)	↑ cAMP → protein kinase A → relaxation

Purinergic contraction and DHIC

- Atropine-resistant (purinergic, P2X1) contraction is **minimal in normal human bladder** but appears in obstructed, OAB and ageing bladders.
- Purinergic activation suppresses muscarinic coupling → explains **detrusor hyperactivity with impaired contractility (DHIC)**: ↑ afferent drive (urgency) + ↓ muscarinic contraction (incomplete emptying).

Gap junctions and myogenic activity

- Cx43 (and Cx45) gap junctions couple cells within bundles, but the detrusor is deliberately **poorly coupled** overall — high tissue impedance, no fused tetanic contractions — which protects against synchronous activation during filling.
- ↑ Cx43 coupling (BOO, NDO, urgency; induced by cytokines and nerve growth factor, NGF) → coordinated whole-bladder myogenic contraction = DO.
- **Interstitial cells**: intradetrusor c-KIT+ cells (modulate Ca^{2+} transients; imatinib ↓ spontaneous contractions; ↑ in OAB/BOO) and inhibitory PDGFRα+ cells (P2Y1-mediated relaxation).

> **KEY POINT** **Motor–sensory coupling:** afferents lie beside myocytes; spontaneous micromotions generate afferent bursts perceived as urgency. Antimuscarinics and mirabegron reduce non-voiding contractions and hence sensory input — efferent and afferent functions are inseparable.

5 · RECEPTORS AND DRUG TARGETS

Receptor	Location	Function	Drug target
M3 (Gq)	Detrusor; urothelium; salivary glands	**Mediates contraction** (knockout proves it)	Antimuscarinics (OAB)
M2 (Gi)	Detrusor (**numerically predominant**); urothelium	Amplifies M3: opposes β-cAMP relaxation, inactivates K^+ channels, opens cation channels; shift M3 → M2 in neurogenic/obstructed bladder	Non-selective antimuscarinics
β3 (Gs)	Detrusor (**most expressed adrenoceptor mRNA**); sacral cord; afferents	Relaxation via cAMP during storage; ↑ compliance, ↓ non-voiding contractions	Mirabegron, †vibegron (OAB)
α1A	Urethra, prostate (dominant)	Urethral contraction / closure; maintains resting female urethral tone	α-blockers ↑ flow in benign prostatic hyperplasia (BPH); α-agonists ↑ closure
α1D	Bladder (normal human predominant α); spinal cord	Storage symptoms of obstruction	α1D-active blockers (tamsulosin) improve storage LUTS
α2	Prejunctional urethra; vesical ganglia	↓ noradrenaline release, ↓ ganglionic transmission	—
P2X3 (P2X2/3)	Suburothelial unmyelinated afferents	Senses urothelial ATP → fullness, pain; ↑ in BPS/IC	P2X3 antagonist AF-219 (BPS/IC proof-of-concept)
P2X1	Detrusor, vascular muscle	Purinergic contraction; ↑ in BOO, ageing	Potential for DHIC
TRPV1	C-fibre nociceptors; urothelium	Heat >43 °C, protons, capsaicin; inflammatory hypersensitivity; ↑ in NDO	Capsaicin, resiniferatoxin (RTX) desensitise
TRPM8	Afferents	Cold → bladder cooling reflex	Experimental
NK1 (substance P)	Spinal cord, vessels	Afferent transmission, neurogenic inflammation	Aprepitant ↓ urgency in OAB (trial)
Nicotinic	EUS (pudendal)	Striated sphincter contraction	—

Antimuscarinics

- **Tissue selectivity matters more than subtype selectivity** — salivary glands also use M3, so M3-selective darifenacin (~30-fold M3 > M2) still causes dry mouth. Tolterodine and solifenacin are relatively bladder-selective.

- **Oxybutynin**: non-selective; its metabolite **desethyloxybutynin** causes most dry mouth → controlled-release oxybutynin keeps efficacy with fewer side effects.
- **Cognition**: lipophilic, blood–brain barrier-crossing agents block CNS M1 → confusion, delirium, cognitive decline; additive anticholinergic burden. Caution in frail elderly/dementia; effects reverse within weeks of withdrawal.

Other storage drugs (Table 103.4)

- Bladder-acting: antimuscarinics, direct relaxants, Ca^{2+} antagonists, K^+ channel openers, β-agonists, tricyclics (imipramine: anticholinergic + direct relaxant + noradrenaline reuptake inhibition), afferent inhibitors (capsaicin, RTX).
- Outlet-acting: **α-agonists** (ephedrine, midodrine) ↑ urethral closure; **oestrogen** ↑ mucosal seal, blood flow and urethral adrenergic response — but alone disappointing for SUI/urgency urinary incontinence (UUI); systemic oestrogen may worsen UI.
- **Duloxetine** (serotonin–noradrenaline reuptake inhibitor): ↑ Onuf's nucleus output → ↑ striated sphincter activity; beats placebo in mild–moderate SUI; approved in Europe, not USA.

6 · PERIPHERAL INNERVATION

Feature	Parasympathetic	Sympathetic	Somatic
Origin	**S2–S4** sacral parasympathetic nucleus (intermediate grey)	Thoracolumbar intermediolateral cell column (†**T10–L2**)	**Onuf's nucleus**, S2–S4 ventral horn (lateral border)
Nerve	Pelvic	Hypogastric	Pudendal
Bladder body	Contracts (ACh → M3; ATP → P2X1)	Relaxes (noradrenaline → β3)	—
Bladder neck / urethra	Relaxes urethra (**NO**)	Contracts (α1)	Contracts EUS (ACh → nicotinic)
Ganglia	Pelvic plexus + **intramural**	Chain → inferior mesenteric → pelvic	None
Net role	**Voiding**	**Storage**	Storage (voluntary closure)

- All three nerves are mixed afferent + efferent.
- Postganglionic parasympathetic neurons lie partly in the bladder wall, so cauda equina or pelvic plexus injury **decentralises rather than fully denervates** the bladder.

Afferents

- Pelvic and pudendal afferents → sacral dorsal root ganglia (DRG); hypogastric → rostral lumbar DRG. Lamina propria plexus densest at bladder neck/proximal urethra, absent at the dome.
- Filling gives episodic sensory bursts of rising intensity; the last is perceived as urgency. Afferent firing during contraction reinforces voiding.

Feature	Aδ (thin myelinated)	C (unmyelinated)
Location	Smooth muscle	Mucosa (± muscle)
Normal role	Low-threshold **tension receptors** — fullness; drive normal reflex	Majority; many "**silent**" to normal distension; nociception
Reflex	Spinobulbospinal (via pons)	New **spinal reflex** after spinal cord injury (SCI), inflammation, BOO, ageing
Capsaicin / RTX	Resistant	Sensitive (blocked)

- **Sensitisation** (leftward shift of stimulus–response curve) by ATP, NGF (→ TrkA), PGs, bradykinin, high K^+, low pH, high osmolality; persistent injury → central sensitisation.

- **Cross-organ sensitisation** (shared DRG/spinal convergence) explains BPS/IC–irritable bowel overlap.
- μ-opioids ↓ bladder sensation and ↑ sphincter tone → postoperative retention.

7 · CENTRAL CONTROL

Spinal transmitters

- **Glutamate**: main excitatory transmitter of the reflex.
- **GABA, glycine**: inhibit micturition (mediate rectovesical inhibition).
- **Serotonin**: inhibits parasympathetic drive, excites sphincter. **Dopamine** tonically inhibits the PMC — lost in Parkinson disease → overactivity.

Brainstem switch: PAG and PMC

- **Pontine micturition centre (PMC) = Barrington's nucleus** = "M region" (dorsal pontine tegmentum). Stimulation → bladder contraction + sphincter relaxation; lesions (stroke, MS, tumours) → **retention**.
- PMC does **not** project to Onuf's nucleus; it relaxes the sphincter **indirectly** via GABA/glycine inhibitory interneurons in the dorsal grey commissure — the command centre for emptying.
- **L-region** (ventrolateral pons) = pontine continence/storage centre; projects to Onuf's nucleus.
- **Periaqueductal grey (PAG)** receives ascending bladder afferents and relays to the PMC; together PAG + PMC form a binary "on–off" switch. Without higher control, the bladder empties involuntarily at a critical volume (infant).

Forebrain

- **Storage sequence**: afferents → PAG (switch not tripped) → thalamus → **insula** (desire to void) → lateral/medial **prefrontal cortex** (decision, social appropriateness) → medial frontal → brainstem pathway **tonically suppresses the reflex** = normal continence mechanism.
- Frontal white-matter (medial prefrontal–PAG) lesions → **incontinence**; dorsal anterior cingulate gyrus and supplementary motor area over-respond in urge incontinence (urgency = emergency "backup" mechanism).

> **KEY POINT** **Lesion localisation logic:** forebrain lesion → loss of tonic inhibition → DO with preserved coordination; pontine lesion → retention; suprasacral cord lesion → loss of PMC coordination → DO + **detrusor–sphincter dyssynergia (DSD)**; sacral/cauda lesion → decentralised, acontractile bladder → *Ch 104*.

8 · STORAGE AND EMPTYING PHASES

Phase	Afferent	Efferent	Central pathway
Storage	Low-level pelvic afferent activity	EUS contraction (somatic); internal sphincter contraction, detrusor and ganglionic inhibition (sympathetic); parasympathetic silent	**Spinal** reflexes
Micturition	High-level pelvic afferent activity	EUS inhibited; sympathetic inhibited; parasympathetic to bladder (contract) and urethra (NO relax)	**Spinobulbospinal** reflex via PAG–PMC

Storage

- Low constant pressure depends on: intrinsic muscle/stroma properties, quiescent parasympathetic outflow, urothelial apical membrane changes, and the **sympathetic storage reflex** (lumbosacral; closes outlet, inhibits detrusor). This reflex is weak in humans — bilateral retroperitoneal lymph node dissection does not alter storage.
- **Guarding reflex**: rising bladder afferent activity recruits pudendal motoneurons → EUS electromyography (EMG) increases with filling. Also triggered by sudden abdominal pressure

(cough) — urethral pressure rises **before** pressure transmission, proving an active neural component. Its suppression during voiding is supraspinal → lost after SCI → **DSD**.

- **Sphincter-to-bladder reflex**: pudendal afferents (sphincter, perineum, anal canal, vagina, penis) inhibit bladder preganglionic neurons and interneurons — basis of "holding on" by pelvic floor contraction and of neuromodulation.

Emptying

ALGORITHM
Tension-receptor (Aδ) firing exceeds threshold → PAG/PMC switch flips (if cortex permits) → sympathetic and pudendal outflow off → **urethral sphincter relaxes first** (seconds earlier) + urethral smooth muscle relaxes (NO) → detrusor contracts (parasympathetic) → flow → urethral flow afferents reinforce contraction until empty

- **Urethra-to-bladder reflexes** (Barrington): flow facilitates bladder contraction — a somatic pudendal component via the PMC and a visceral pelvic spinal component. Explains a link between stress incontinence (urine entering the proximal urethra) and detrusor instability.
- After SCI: spinal shock (retention for weeks) → a **C-fibre spinal micturition reflex** emerges via synaptic reorganisation, driven by NGF → NDO with DSD; reciprocity is lost. Ice-water test = research tool only.

Voiding mechanics

- **Pdet = Pves − Pabd**. Detrusor power ≈ Pdet × flow (Q) is roughly constant, so voiding **Pdet reports outlet resistance, not contractile strength**.
- SUI women void at Pdet at Qmax ≈ **19 cmH_2O**; healthy men ≈ **60–70 cmH_2O**. Low voiding pressure in a woman ≠ impaired contractility; male obstruction nomograms must not be applied to women.

9 · URETHRAL CLOSURE MECHANISMS

- **Contributors to urethral pressure**: smooth muscle, striated muscle and lamina propria/vascular plexus. Nicotinic blockade rarely lowers tone >40%; α-blockade lowers urethral pressure by about **one third** → smooth muscle (α1A) tone matters; little cholinergic contribution.
- EUS pressure rises uniformly like an **iris** despite its posteriorly open horseshoe; noradrenaline augments it.
- **Mucosal seal**: vascular filling of lamina propria; oestrogen ↑ urethral blood flow. Impaired arterial supply ↓ urethral pressure (early vascular, later hypoxic muscle effect).
- **Passive pressure transmission** to the proximal urethra during raised abdominal pressure.
- **Hammock hypothesis (DeLancey)**: abdominal pressure compresses the urethra against a posterior layer kept firm by pelvic muscle/fascial support (pubourethral and vaginal–fascial attachments).
- **Active** guarding reflex: fast-twitch fibres add a reflex squeeze on coughing.

> **KEY POINT** Continence = **active muscle tone + passive anatomical coaptation**. The integral theory led to the mid-urethral sling (MUS), the surgical standard for SUI, yet DeLancey's data suggest intrinsic sphincter function matters more than support → *Ch 109*.

10 · COMPLIANCE

- **Compliance C = ΔV / ΔP**. Physiological filling keeps pressure **<10 cmH_2O**; women with SUI filled at 50 mL/min rose only a few cmH_2O to capacity.
- Determinants: (1) passive collagen, elastin, muscle viscoelasticity; (2) active muscle tone; (3) geometry. During filling the lamina propria thins, bundles realign side-to-side and coiled **type III collagen straightens** parallel to the lumen.

- **Low compliance** (steep curve): fast filling, ↑ collagen/↓ elastin, smooth muscle hyperactivity, or combinations.
- Compliance is **fill-rate dependent**: pressure falls when urodynamic filling slows or stops — so slow the pump before diagnosing low compliance.
- Active component: acute spinal transection does not alter compliance, but pelvic nerve transection reduces it → an afferent–CNS loop inhibits spontaneous contractility during filling.
- † A detrusor leak point pressure ≥ **40 cmH_2O** in low-compliance neurogenic bladders predicts upper tract damage — compliance is the main renal-risk parameter → *Ch 104*.

11 · PATHOPHYSIOLOGY LINKS

Idiopathic OAB / DO

- DO is found in only a minority (11% in one OAB cohort) — symptoms reflect dysregulated sensation more than contraction → "afferent neurourology".
- Mechanisms: ↑ urothelial ATP/TRPV1, ↑ Cx43 coupling, ↓ BK, NGF-driven plasticity.

Bladder outlet obstruction (BOO)

- Detrusor hypertrophy (no mitoses), partial denervation → **muscarinic supersensitivity**, ↑ P2X1 and Cx43 → coordinated myogenic contraction (myogenic theory, Brading).
- Neural: NGF from strained DSM enlarges afferent/efferent neurons and builds new spinal circuits; NGF stays ↑ when frequency persists after relief — explaining **persistent storage symptoms after prostate surgery**.

BPS/IC

- Afferent hyperexcitability: ↑ NGF, ATP/P2X3, TRPV1, mast cell activation. Single-target trials (anti-NGF tanezumab, intravesical RTX) failed — likely patient heterogeneity → *Ch 113*.

Underactive bladder

- UAB = symptom complex (weak stream, hesitancy, straining); detrusor underactivity = urodynamic finding. Afferent loss may cause underactivity, not just efferent failure.
- Post-void residual (PVR) ≥ **300 mL** warrants further evaluation (also if bothersome LUTS, hydronephrosis, renal insufficiency, culture-proven UTI); PVR <300 mL does not prove normal contractility.

12 · AGEING

- Ageing = **hypofunction**: detrusor contractility, bladder sensation and urethral closure pressure all decline. Capacity does not fall with age per se — smaller capacity reflects coexisting DO.
- Structure: ↓ smooth muscle relative to collagen; pelvic ischaemia → LUTS, DO.
- Nerves: ↓ cholinergic nerves, ↓ afferent neuropeptides, ↓ brain response to filling.
- Receptors: cholinergic response unchanged (despite ↓ muscarinic density); **α-contraction ↑ (α1D) and β-relaxation ↓** → noradrenaline contracts the aged bladder but relaxes the young one.
- Aged CNS is prone to overactivity despite ↓ baseline function → urgency with poor emptying (DHIC).
- Antimuscarinics: anticholinergic burden linked with dementia risk (causality unclear) → †prefer β3 agonists or low-CNS-penetrant agents in the elderly.

13 · NEUROMODULATION AND BOTULINUM TOXIN — MECHANISMS

- **Sacral neuromodulation (SNM)**, pudendal and posterior tibial nerve stimulation (PTNS) treat refractory OAB **and** non-obstructive urinary retention (NOUR). Target = **afferent axons**.
- **Paradox resolved**: in OAB, somatic afferent input inhibits bladder preganglionic neurons and interneuronal transfer to the PMC; in retention, it switches off the overactive **guarding reflex**, releasing the bladder.

- Pudendal stimulation carries S2–S4 afferents and avoids sciatic-territory discomfort of S3 stimulation.
- **OnabotulinumtoxinA (OBTX)**: heavy chain binds nerve terminal → internalisation → light chain (zinc protease) cleaves **SNAP-25** → no vesicle fusion → ↓ ACh release. Also afferent: ↓ TRPV1/P2X3 in C fibres, ↓ SP, CGRP and urothelial ATP.

14 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ABC	RCT: intradetrusor OBTX vs oral antimuscarinic, idiopathic OAB with UUI	Equal ↓ in daily UUI; OBTX more complete dryness	Antimuscarinic: dry mouth; OBTX: catheterisation, UTI
ROSETTA	RCT: OBTX 200 U vs SNM, women with refractory non-neurogenic OAB, 2 years	Comparable efficacy; OBTX more often dry	OBTX more UTIs (note 200 U dose)

15 · MCQ ONE-LINERS

1. Muscarinic subtype most numerous in detrusor: **M2**; subtype mediating contraction: **M3**.
2. Most highly expressed adrenoceptor in human detrusor: **β3**.
3. Dominant α-subtype in urethra and prostate: **α1A**; in human bladder: **α1D**.
4. Main Ca^{2+} source triggering detrusor contraction: **L-type VDCC**.
5. Calcium sensitisation: **Rho kinase** inhibits MLCP.
6. Key K^+ channel braking detrusor excitability: **BK** (large-conductance Ca^{2+}-activated).
7. Uroplakins bind **type 1 fimbriae** of uropathogenic *E. coli*.
8. Urothelial transmitter exciting afferents via P2X3: **ATP**; inhibiting them: **NO**.
9. Parasympathetic transmitter relaxing urethra during voiding: **NO**.
10. Somatic EUS motoneurons: **Onuf's nucleus** (S2–S4 ventral horn).
11. PMC = **Barrington's nucleus** (dorsal pontine tegmentum); lesion → retention.
12. Pontine storage centre projecting to Onuf's nucleus: **L-region**.
13. Normal micturition reflex: **spinobulbospinal**, Aδ afferents.
14. Reflex emerging after SCI: **C-fibre spinal**, capsaicin/RTX-sensitive.
15. Dopamine tonically **inhibits** PMC — lost in Parkinson disease.
16. Loss of supraspinal suppression of guarding reflex → **DSD**.
17. Female rhabdosphincter: **87%** slow-twitch fibres.
18. Collagen type ↑ in poorly compliant bladder: **type III**.
19. α-blockade lowers urethral pressure by about **one third**.
20. Oxybutynin dry mouth mainly from metabolite **desethyloxybutynin**.
21. OBTX cleaves **SNAP-25**.
22. Duloxetine acts on **Onuf's nucleus** to ↑ sphincter tone.
23. Aged bladder: α-contraction ↑, β-relaxation ↓.
24. Neurotrophin central to DO plasticity: **NGF**.
25. RTX desensitises TRPV1 **without prior excitation**, unlike capsaicin.

104 Pathophysiology and Classification of Lower Urinary Tract Dysfunction

Reading time ≈ 16 min (slow pace)

1 · NORMAL FUNCTION: TWO PHASES

The lower urinary tract (LUT) stores at low pressure with continence and empties completely, voluntarily, at low pressure. Phases switch on–off: storage reflexes on/voiding off ⇄ the reverse.

Storage requires	Voiding requires
Low detrusor pressure (normal compliance), appropriate sensation	Detrusor contraction of adequate magnitude and duration
Outlet closed at rest and on straining	Fall in smooth and striated sphincter resistance
No detrusor overactivity (DO)	No anatomic obstruction

- **Storage is mostly passive**: compliance (Δvolume/Δpressure) from elastic/viscoelastic stroma. ↑ collagen (inflammation, bladder outlet obstruction [BOO], decentralisation) → low compliance, then **drug-unresponsive**.
- Storage reflexes: α1 → bladder neck tone; β3 → detrusor relaxation; spinal **guarding reflex** raises urethral pressure.
- **Voiding**: pelvic-nerve parasympathetic outflow, coordinated by the **pons**; outlet relaxes first (nitric oxide), then detrusor contracts.
- **Stress continence**: coughing does not open the outlet because no coordinated contraction is triggered. Protection = at least equal pressure transmission to proximal urethra (**Enhorning**) plus active reflex closure (**Tanagho**). Its failure accompanies all effort-related urinary incontinence (UI).
- Urethral closure also needs a **mucosal seal** — scarred or atrophic mucosa seals poorly. **Klevmark**: the cystometric pressure rise is an artefact of fast filling.

2 · PATHOPHYSIOLOGY

Excluding psychological causes, every failure is due to the **bladder, the outlet, or both** — the basis of the **Wein functional classification**. Failure is usually relative; combinations are common.

Storage failure

- **Bladder overactivity** = involuntary contractions and/or low compliance: neurological disease, BOO, ageing, ↑ afferent input, idiopathic.
- **Overactive bladder (OAB)** (International Continence Society, ICS): urgency ± urgency urinary incontinence (UUI), usually with frequency and nocturia. UUI → assume DO; urodynamic studies (UDS) often miss it. Mechanisms: ↓ suprapontine inhibition, axonal damage, enhanced excitatory transmission, ↑ afferent input. Pelvic floor contraction reflexly inhibits the detrusor — basis of bladder training.
- **Hypersensitivity** without overactivity: classically interstitial cystitis/bladder pain syndrome (IC/BPS).
- **Low compliance**: sacral/infrasacral injury, fibrosis from long-standing BOO.
- **Outlet underactivity**: sphincter denervation, surgery/trauma, ageing, loss of female support.

	Hypermobility (genuine stress urinary incontinence, SUI)	Intrinsic sphincter deficiency (ISD)
At rest	Outlet competent	Bladder neck poorly functional/open
Cause	Poor pelvic support	Prior surgery, scarring, neurological
Treatment	Correct hypermobility	Coaptation/compression

KEY POINT UI can occur from ISD alone, **never from hypermobility alone**. **DeLancey's hammock**: a firm suburethral layer lets abdominal pressure compress the urethra. In half of continent women the continence point is the **mid-urethra**, not the bladder neck. Male sphincteric incontinence resembles ISD, not hypermobility.

Emptying failure

- **Bladder underactivity**: neurogenic (incl. pelvic surgery), myogenic (overdistension, drugs, infection), psychogenic; diabetes = neuromuscular + vascular.
- **BOO** (ICS): low flow and/or raised post-void residual (PVR) **with high detrusor pressure**. Anatomic — male: prostate, bladder neck contracture, stricture; female: prolapse, post-sling.
- **Functional**: striated sphincter dyssynergia (neurological, commonest); smooth sphincter dyssynergia (autonomic dysreflexia); primary bladder neck obstruction (young men); dysfunctional voiding (non-neurogenic).

3 · FUNCTIONAL CLASSIFICATION (WEIN) — 2 × 2 TABLE

	Because of the bladder	Because of the outlet
Failure to store	DO; low compliance; hypersensitivity; underactivity with overflow	Hypermobility; ISD. (Plus **fistula**)
Treatment	Bladder training; antimuscarinics, β3-agonists, onabotulinumtoxinA; neuromodulation; augmentation	Pelvic floor training; α-agonists, duloxetine; bulking; slings; artificial urinary sphincter
Failure to empty	Underactivity: neurogenic, myogenic, psychogenic	Anatomic obstruction; sphincter dyssynergia; bladder neck dysfunction
Treatment	Valsalva/Credé; parasympathomimetics; electrical stimulation	α-blockers, 5α-reductase inhibitors; prostatectomy, incision; sphincter onabotulinumtoxinA, sphincterotomy
Circumvent	Pads/collecting devices, antidiuretics, catheterisation, diversion	

KEY POINT

- Bladder underactivity appears under **both** storage (overflow) and emptying. MCQ
- α-agonist tightens outlet (storage); α-blocker relaxes it (emptying).
- OnabotulinumtoxinA: intradetrusor → storage; sphincteric → emptying.
- Naming the failure names the treatment family — no exact diagnosis needed.

Combined deficits (T10 paraplegia: DO + sphincter dyssynergia): treat one as primary — circumvent emptying with external collection, or **abolish DO** and empty by clean intermittent catheterisation (CIC). Other combinations: BOO with DO; DO with impaired contractility. Chronic obstruction causes hypertrophy, collagen, diverticula and DO — intervene promptly.

4 · URODYNAMICS BY PHASE AND SITE

Phase	Bladder	Outlet
Storage	Filling cystometry; detrusor leak point pressure (DLPP)	Urethral pressure profile; Valsalva leak point pressure (VLPP)
Voiding	Voiding cystometry (Pdet)	Fluoroscopy; sphincter electromyography
Both	Flowmetry and PVR — **cannot localise** the fault	

5 · OTHER CLASSIFICATIONS

- **Symptoms (ICS)** — patient report, directs testing. Storage: frequency, nocturia, urgency, UI. Voiding: slow stream, intermittency, hesitancy, straining, **terminal dribble**. Post-micturition: incomplete emptying, **post-micturition dribble**.

- **Signs (ICS)**: bladder diary, examination (SUI, prolapse may mask UI), pad test.

ICS urodynamic classification (most precise)

Domain	Storage	Voiding
Detrusor	Normal; DO (neurogenic/idiopathic)	Normal; underactive; acontractile
Sensation, capacity, compliance	Normal, high/increased, low/reduced, absent	—
Urethra	Normal or incompetent closure	Normal; BOO, dysfunctional voiding, detrusor–sphincter dyssynergia (DSD)

- Cystometric capacity = permission to void given; maximum cystometric capacity = can no longer delay; maximum anaesthetic capacity = filled under anaesthesia.
- DSD = detrusor contraction with concurrent involuntary striated sphincter contraction; prefix "neurogenic" when a relevant neurological disorder exists.

Contrast: T10 paraplegia — neurogenic DO, absent sensation, DSD; stroke — neurogenic DO, normal sensation, **synergic** voiding.

† **Lapides**: sensory neurogenic, motor paralytic, uninhibited, reflex, autonomous bladder. **Bors–Comarr**: upper/lower motor neuron, complete/incomplete, balanced/unbalanced. MCQ

6 · NOCTURIA AND NEUROGENIC DYSFUNCTION

- **Polyuria >40 mL/kg/24 h**. **Nocturnal polyuria**: night volume ÷ 24-h volume **>20% young, >33% elderly**; from loss of nocturnal vasopressin rise or obstructive sleep apnoea.
- **IC/BPS**: urgency is driven by pain/pressure worsening with filling. Normally silent unmyelinated **C fibres** become hyperactive (central sensitisation); myelinated **Aδ fibres** normally signal distension. Bladder–bowel crosstalk explains OAB–irritable bowel overlap.
- **Neurogenic LUT dysfunction** (ICS) requires a clinically confirmed relevant neurological disorder.

Lesion	Detrusor	Sphincter
Suprapontine	DO	Synergic
Suprasacral spinal	DO	DSD, raised PVR
Sacral	Acontractile	Impaired

> **KEY POINT** DSD needs an intact sacral arc **disconnected from the pons** — hence suprasacral, never suprapontine.

7 · MCQ ONE-LINERS

1. Wein: failure to store or empty, due to bladder or outlet.
2. Storage failure outside bladder and outlet: fistula.
3. Micturition coordinating centre: pons.
4. Stress continence: equal pressure transmission to proximal urethra (Enhorning).
5. Collagen-driven low compliance is drug-unresponsive.
6. Flow and PVR cannot distinguish weak detrusor from obstruction.
7. DLPP assesses bladder; VLPP assesses outlet.
8. BOO: high detrusor pressure with low flow.
9. Incontinence never results from hypermobility alone.
10. Smooth sphincter dyssynergia: autonomic dysreflexia.
11. Dysfunctional voiding is non-neurogenic.
12. Polyuria: >40 mL/kg/24 h.
13. Nocturnal polyuria index >33% in elderly.
14. Terminal dribble: voiding symptom; post-micturition dribble: post-micturition.
15. Suprasacral lesion: DO + DSD.
16. Sacral lesion: acontractile detrusor.
17. Suprapontine lesion: synergic voiding.
18. Maximum cystometric capacity: can no longer delay voiding.
19. Normally silent bladder afferents: C fibres.

10. Bladder underactivity causes overflow incontinence too.

107 Urodynamic and Videourodynamic Evaluation of the Lower Urinary Tract

Reading time ≈ 41 min (slow pace)

1 · DEFINITION AND ROLE

- **Urodynamics (UDS)**: dynamic study of transport, storage and evacuation of urine — tests used singly or together to diagnose suspected lower urinary tract (LUT) dysfunction and assess the effect of known dysfunction. International Continence Society (ICS) separates UDS *studies* (tests) from UDS *observations* (what is seen).
- LUT has two jobs: **low-pressure storage** (protects kidneys, continence) and **voluntary evacuation**. Failure gives bothersome lower urinary tract symptoms (LUTS) or harm (urinary tract infection (UTI), renal failure) — sometimes silently.
- UDS is useful when history, examination and simple tests are insufficient: (1) to **diagnose the cause of symptoms**; (2) to determine the **impact of a disease that can silently damage the upper tracts** — spinal cord injury (SCI), multiple sclerosis (MS), radiation cystitis.
- Roles: show pathophysiology, predict upper-tract risk and treatment outcome, explain treatment failure.

Wein functional classification — the frame for every study

Site	Overactive	Underactive
Bladder	Failure to store	Failure to empty
Outlet	Failure to empty	Failure to store

Each study is designed to place the patient in this grid: bladder, outlet or both — storage, emptying or both.

Principles of interpretation

- Decide the questions first → design the study to answer them → customise. Symptoms alone are not an indication.
- Aim to **reproduce the symptom** and identify its cause. An abnormality without symptom reproduction is not necessarily diagnostic; failure to record an abnormality does not exclude it.
- Interpret with history, postvoid residual (PVR) and frequency–volume chart.

> KEY POINT **Box 107.1 — Urodynamic risk factors** (treat even if asymptomatic): impaired compliance; detrusor–external sphincter dyssynergia (DESD); detrusor–internal sphincter dyssynergia (DISD); high-pressure detrusor overactivity throughout filling; detrusor leak point pressure **> 40 cmH_2O**; poor emptying with high storage pressure. Each is a route to sustained high intravesical pressure.

2 · PREPARATION

Clinician and patient

- **Medication status** (antimuscarinics, β_3-agonists) decided and documented in advance: to assess drug effect → test on regular dosing; to uncover aetiology of empirically treated symptoms → stop the drug first.
- **ICS pre-invasive work-up**: validated symptom/bother score, **3-day frequency–volume chart**, free uroflowmetry and PVR. A normal free flow and low PVR beforehand let an abnormal invasive flow be recognised as artefact.
- Explain purpose; well tolerated; commonest complaint afterwards is dysuria.
- **Urinalysis before every invasive study**; symptomatic UTI → postpone until treated.

Antibiotic prophylaxis (AUA / SUFU Best Practice Policy)

"Index" patient (normal anatomy, no risk factors): **no routine prophylaxis** (level 1). Non-index: **single dose within 1 h** — first choice oral trimethoprim–sulfamethoxazole; alternatives cephalosporin, amoxicillin–clavulanate, aminoglycoside ± ampicillin, fluoroquinolone.

Prophylaxis YES	Prophylaxis NO
Neurogenic LUT dysfunction (NLUTD); bladder outlet obstruction (BOO) or raised PVR; age > 70; **asymptomatic bacteriuria**; immunosuppression (esp. renal transplant); chronic catheters incl. clean intermittent catheterisation (CIC); high-risk orthopaedic implant	Diabetes; **recurrent UTI history**; recent hospitalisation; obesity; menopause; gender; prior urological surgery; cardiac valve disease (not routinely)

Pattern: "yes" = stasis, colonisation, foreign material, immune compromise, implants. Trap: asymptomatic bacteriuria is treated here, unlike most urological settings.

Facility

- Quiet private room with space for wheelchairs/hoists; videourodynamics (VUDS) needs fluoroscopy space. Have pessary/Scopette for prolapse reduction.
- Unanswered question → repeat fill/void in the same session.

3 · NON-INVASIVE COMPONENTS

Bladder diary (frequency–volume chart)

- 3 days recommended; gives functional capacity, voiding frequency, nocturnal output. Used to set the cystometric target volume and to judge whether a flow's voided volume is representative.

Uroflowmetry

- Parameters: voided volume (VV), maximum and average flow (Qmax, Qave), voiding time vs flow time, time to Qmax; **voiding efficiency = VV / (VV + PVR) × 100**.
- **Qmax depends on VV**: linear above 150 mL, hyperbolic below → minimum **VV 150 mL**. Alternatives: corrected Qmax (Qmax/√VV), Siroky nomogram (men), Liverpool nomogram (men and women).
- † Usual norms: Qmax > 15 mL/s in men (< 10 suggests obstruction), > 20 mL/s in women.
- Curve: **bell-shaped** normal; **flattened/plateau** → fixed obstruction (e.g. stricture); **interrupted/straining** → underactivity, obstruction or abdominal voiding.
- Limitation: flow cannot say *why* — low flow is obstruction or weak detrusor; only detrusor pressure separates them.

Postvoid residual

- By bladder scan or catheter (state method). Shows emptying is impaired, not why. Confirm a raised PVR with a repeat, especially if there was no true desire to void. No agreed "elevated" threshold.

4 · EQUIPMENT, TRANSDUCERS AND QUALITY CONTROL

- Minimum: multichannel — vesical pressure (Pves), abdominal pressure (Pabd), subtracted detrusor pressure (Pdet) and flow; optional electromyography (EMG), urethral pressure. Infusate: saline or contrast; **not CO_2**.

Feature	External water-filled	Air-charged	Microtip
Reference level	**Upper border of symphysis pubis**; re-level on position change	None (no hydrostatic effect)	—
Problems	Air bubbles; motion and hydrostatic artefact; underdamped	Overdamped (> 3 Hz attenuated) — blunts coughs	Expensive, reusable
Standards	Basis of most nomograms (reference standard)	No reference values	—

- Systems are **not interchangeable** — Pdet may differ by up to 10 cmH_2O; patterns transfer, absolute values (pressure–flow, urethral profile) do not.

> **KEY POINT** **Quality control** † (ICS Good Urodynamic Practice): zero to atmospheric pressure; flush lines free of air; check initial resting Pves and Pabd are physiological and similar (Pdet near 0); **cough test** before filling, at intervals and before voiding — equal Pves and Pabd spikes confirm transmission (a biphasic Pdet spike means poor transmission → re-flush/reposition). Common artefacts: rectal contractions (fall in Pdet without cause), catheter displacement or loss of signal, kinked tubing, flow artefact from stream hitting the funnel edge or tapping.

5 · FILLING CYSTOMETRY (CMG)

Technique

- Filling runs from start of infusion to "permission to void" = **maximum cystometric capacity (MCC)** — a strong but not uncomfortable desire; set target from diary and prior PVR.
- **Pdet = Pves − Pabd**. Single-channel Pves cannot tell detrusor from abdominal rises; Pabd by rectal catheter preferred (vaginal/stomal less validated).
- Filling rates: standard adult **30–100 mL/min** (medium); † ICS physiological rate < body weight (kg)/4 mL/min; slow in neurogenic bladders. Neurogenic: physiological rate, body-warm saline (EAU).
- Position that provokes the symptom — DO is commoner upright or on moving from supine.

What is measured

- **Normal (formerly "stable")**: Pdet near zero throughout, no involuntary contractions.
- **Sensation** — the only subjective parameter: † first sensation, first desire, strong desire, urgency.
- **Capacity**, **compliance**, **involuntary detrusor contractions (IDC)**.

Detrusor overactivity (DO)

- Urodynamic observation of IDCs during filling, spontaneous or provoked. **Idiopathic DO (IDO)**; **neurogenic DO (NDO)** — relevant neurological disorder; non-neurogenic secondary DO. Terms "hyperreflexia"/"instability" obsolete. Label comes from history — tracings look identical.
- Absence does not exclude it: **up to 50%** of women with urgency urinary incontinence (UUI) show no DO. DO occurs in a minority of asymptomatic volunteers, more on ambulatory UDS.
- Confirm it is involuntary (confused patient may simply void); record volume, pressure, leakage (DO incontinence); repeat at slower fill if uncharacteristic. Characterise: single/multiple, phasic/terminal, suppressible or not.
- Prognosis: in BOO with overactive bladder (OAB), a **single terminal IDC** predicts storage-symptom resolution after transurethral resection of prostate (TURP). In NDO, higher Pdetmax and longer IDCs → upper-tract deterioration.
- **Stress-induced DO** (ICS: cough-associated DO): cough triggers an IDC — mimics stress urinary incontinence (SUI) but is detrusor in origin.

Compliance

- **Compliance = ΔV / ΔPdet** (mL/cmH_2O), between start of filling and MCC (or before a leak-causing contraction), excluding contractions.
- No agreed normal; < 20 mL/cmH_2O proposed as poor. But a small-capacity bladder gives a low number without danger → **absolute storage pressure matters more**: **> 40 cmH_2O** associated with upper-tract damage.
- Causes of impaired compliance: neurological (SCI, spina bifida — outlet resistance from DESD, or lower motor neuron (LMN) decentralisation); long-term BOO; radiation; tuberculosis.

Compliance pitfall	Resolution
Rapid filling → accommodation artefact	Stop/slow filling; pressure falls

Compliance pitfall	Resolution
Sustained low-amplitude IDC mimics low compliance	Stop filling — returns to baseline
Pop-off: vesicoureteric reflux (VUR), diverticulum → compliance looks falsely good	VUDS
Pop-off: incompetent outlet vents pressure	Occlude outlet; may appear only after sling/artificial urinary sphincter

6 · LEAK POINT PRESSURES

Two independent measurements; both seen only in incontinent patients. Record provocation and site.

	Abdominal LPP (ALPP)	Detrusor LPP (DLPP)
Definition	Total intravesical pressure at which leakage occurs from raised abdominal pressure, no detrusor contraction	Lowest Pdet at which leakage occurs without detrusor contraction or abdominal strain
Measures	Sphincter strength	Bladder's response to outlet resistance
Bad value	**Low** = weak sphincter	**High** = danger to upper tracts
Setting	SUI	Low compliance / neurogenic bladder

ALPP technique

- Standing; total pressure including baseline; start at **150 mL**, repeat every 50 mL; no leak at capacity → remove urethral catheter and read from rectal line. Valsalva LPP (VLPP) is lower than cough LPP — state method. Larger catheter → lower ALPP; some women and more post-prostatectomy men leak only with catheter out.

ALPP (cmH_2O)	Inference (McGuire)
< 60	Intrinsic sphincter deficiency (ISD)
60–90	Equivocal
≥ 90	Little/no ISD (hypermobility)

Caution: ISD and hypermobility (Q-tip > 30°) coexist; no hypermobility → SUI must be ISD; ALPP alone poorly predicts surgical success. ICS umbrella term: **urodynamic stress test**.

DLPP

- McGuire: myelodysplastic children with **DLPP > 40 cmH_2O** → hydronephrosis or VUR in 85%. Not a validated safe cut-off (20 cmH_2O more sensitive in children) — target **as low as achievable**.
- Why it matters: bladder pressure is already dangerous before urethral "pop-off". High outlet resistance → less leakage but transmission to kidneys. Treat by lowering bladder pressure (or outlet resistance, e.g. sphincterotomy for DESD) → repeat UDS.

Occult SUI: SUI in a continent woman with pelvic organ prolapse (POP) revealed only on prolapse reduction.

7 · URETHRAL PRESSURE PROFILOMETRY (UPP)

- Brown–Wickham: perfused side-hole catheter withdrawn slowly; records pressure needed to open the urethra.
- Parameters: maximum urethral pressure; **maximum urethral closure pressure (MUCP)** = max urethral − intravesical; functional length (women ~3 cm; MUCP 40–60 cmH_2O normal).
- **MUCP ≤ 20 cmH_2O** → ISD ("type III" SUI), predicted sling/suspension failure.
- Unlike ALPP, UPP does not need or diagnose SUI. ICS: unclear utility — does not discriminate incompetence, grade severity or predict surgical success; EAU: do not use UPP or LPP to grade severity.

8 · PRESSURE–FLOW STUDIES (PFS)

- Voiding from permission to void until the patient considers it finished; simultaneous flow; note inhibition from the setting.

Pdet	Flow	Interpretation
Low/normal	High/normal	Unobstructed
High	Low	**BOO**
Low	Low	Detrusor underactivity (obstruction may be hidden)

- ICS terms: normal detrusor function; **detrusor underactivity** (reduced strength and/or duration → prolonged or incomplete emptying); **acontractile detrusor** ("neurogenic acontractile" replaces "areflexia"); situational inability to void.

Men — nomograms and indices

Abrams–Griffiths, Schäfer linear passive urethral resistance relation (LinPURR, grades obstruction and contractility), ICS provisional nomogram — all derived from TURP patients and classify similarly.

Index	Formula	Cut-offs
Bladder outlet obstruction index (BOOI)	PdetQmax − 2Qmax	**> 40 obstructed · 20–40 equivocal · < 20 unobstructed**
Bladder contractility index (BCI)	PdetQmax + 5Qmax	**> 150 strong · 100–150 normal · < 100 weak**

- Limitation: BOOI ≥ 40 needs Pdet ≥ 40 even at zero flow → a decompensated detrusor masks obstruction.
- **Stop tests** estimate isovolumetric pressure (Piso): voluntary (lowest value), mechanical, continuous occlusion. In men Piso < 50 cmH_2O suggests underactivity.
- BOO also affects storage: ~2/3 of men with benign prostatic obstruction have DO, resolving in half to two-thirds after relief.
- **Micturitional UPP** (Yalla): localises obstruction — drop > 5 cmH_2O within prostatic urethra.
- MCQ Bethanechol supersensitivity test (Lapides; Cannon's law of denervation): 2.5 mg SC; rise **≥ 15 cmH_2O** at 100 mL = positive (neurogenic). Unreliable, rarely changes management (CIC regardless).

Women

- Women void at low pressure (pelvic floor relaxation, some by straining); no benign prostatic hyperplasia (BPH)-like model → **BOOI/ICS nomogram cannot be used** (grossly underestimate female BOO).
- Suspect with LUTS, incomplete emptying, recurrent UTI, prior anti-incontinence surgery, prolapse.

Method	Criterion	Comment
Cut-points	e.g. Qmax ≤ 12 mL/s + PdetQmax ≥ 25 cmH_2O (Defreitas); Chassagne ≤ 15 / ≥ 20	Anatomic obstruction only; overlap
VUDS (Nitti)	Radiographic obstruction between bladder neck and meatus with sustained contraction of any magnitude	Localises; works despite weak detrusor
Blaivas–Groutz	Free Qmax vs Pdetmax; none/mild/moderate/severe	Two separate voids; **overdiagnoses**
Solomon–Greenwell	PdetQmax − 2.2Qmax: < 0 unlikely; > 18 almost certain	Includes functional obstruction; newer

Female underactivity: no universal criteria; associated with reduced sensation.

9 · EMG AND SPHINCTER COORDINATION

- ICS standard: two perineal skin-surface electrodes. **Concentric needle**: better signal, uncomfortable, dislodges.
- Normal: EMG quietens and urethral pressure falls **before** Pdet rises; returns after voiding.
- **Guarding reflex**: EMG rises during an IDC to inhibit it — normal, not dyssynergia.

	DESD	**Dysfunctional voiding**
Cause	**Suprasacral cord lesion** (higher → likelier)	Learned; no neurological lesion
Behaviour	Involuntary sphincter contraction during DO/voiding	Intermittent sphincter contraction during voiding
Significance	High pressure → poor compliance → upper-tract risk	LUTS; confirm intermittent free flow; fluoroscopy improves accuracy

- **DISD**: suprasacral lesion above sympathetic outflow (T10–L1); usually with DESD; **VUDS only**.
- **Primary bladder neck obstruction (PBNO)**: in non-neurogenic men, women, children; bladder neck invisible to EMG → needs fluoroscopy.

10 · VIDEOURODYNAMICS

- Simultaneous UDS + fluoroscopy — most precise evaluation where anatomy and function interact; **gold standard in NLUTD** (EAU: optimum test; if unavailable, CMG + PFS).
- Adds "where": localises obstruction; only way to assess **bladder neck** (PBNO, DISD); confirms equivocal EMG; shows VUR (volume, pressure, grade) and diverticula; shows incontinence not seen on examination; DLPP in patients who cannot be positioned (tetraplegics).
- **Reflux pop-off**: without fluoroscopy compliance is overestimated — a dangerous bladder looks safe.
- Technique: C-arm or tilting table; SUI views standing; void in usual position; screen only at provocation, pressure rises and voiding (lowest dose).
- PBNO in young men: type I high-pressure/low-flow; type II normal pressure, low flow, narrow neck; type III delayed bladder-neck opening — fluoroscopy critical for II and III.
- Refer to VUDS centre: NLUTD, unexplained retention in women, prior radical pelvic surgery, diversion, renal transplant, pelvic radiation; young men/women without anatomical cause (AUA/SUFU).

11 · AMBULATORY URODYNAMICS AND NEWER TESTS

- **Natural filling** with portable recording. Some neuropathic bladders with poor compliance on conventional CMG are normal on ambulatory monitoring — rapid filling overstates pressure. DO is detected more often (including in asymptomatic people).
- † Indication: symptoms (e.g. urgency/UUI) unexplained by repeated conventional UDS.
- Penile cuff test, detrusor wall thickness, near-infrared spectroscopy — none yet replaces subtracted cystometry.

12 · INDICATIONS BY CONDITION (AUA/SUFU, EAU)

Female SUI

- Pure, clinically demonstrated, uncomplicated SUI ("index" patient): UDS may be omitted (AUA/SUFU 2023; EAU: do not routinely perform).
- Perform UDS in non-index: prior anti-incontinence or POP surgery, symptom–examination mismatch, voiding dysfunction, significant urgency/UUI, raised PVR, unconfirmed SUI, NLUTD.
- When SUI not shown with catheter in place → repeat with catheter removed. High-grade POP → stress test with reduction.

Male LUTS

- PVR and uroflow optional/recommended in initial evaluation; no UDS needed to start an α-blocker.
- AUA/SUFU: **PFS when it matters whether obstruction is present, before invasive, morbid or irreversible treatment**. No obstruction → worse surgical results.
- EAU PFS triggers before invasive treatment: prior failed treatment; cannot void > 150 mL; **Qmax > 10 mL/s**; PVR > 300 mL; age > 80 or < 50. Each marks a man in whom obstruction cannot be assumed.
- **Impaired compliance is the only absolute urodynamic indication to treat BOO.**
- OAB: no UDS in uncomplicated initial work-up; consider in refractory/complicated cases before invasive therapy.

Women with voiding symptoms

- PFS separates obstruction from impaired contractility; functional obstruction (PBNO, dysfunctional voiding) is invisible on examination/endoscopy → VUDS.

NLUTD

- Goal: **low-pressure filling** even at the cost of incomplete emptying and catheterisation; only UDS distinguishes DO from poor compliance. RCTs unethical.
- UDS essential before treatment and in follow-up in SCI, myelomeningocele (potentially harmful storage); helpful in BPO with Parkinson disease/MS; usually not needed first in post-stroke UUI.
- Non-invasive testing first (PVR, diary); CMG ± PFS, EMG; VUDS preferred.

13 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ValUE	RCT, 630 women, uncomplicated SUI: office evaluation ± UDS	Success ~77% both arms	Office evaluation alone non-inferior before SUI surgery
Dutch embedded RCT	Discordant UDS randomised: immediate mid-urethral sling vs UDS-tailored	Equal cure	UDS discordance need not change surgery
SISTEr	Burch vs pubovaginal sling	ALPP/DO did not predict outcome	UDS adds little in stress-predominant SUI

14 · MCQ ONE-LINERS

1. Pdet = Pves − Pabd.
2. Water-filled transducer reference level: upper border of symphysis pubis.
3. Minimum voided volume for reliable Qmax: 150 mL.
4. Plateau flow curve: fixed obstruction such as stricture.
5. Standard adult filling rate: 30–100 mL/min.
6. Compliance = ΔV/ΔPdet; < 20 mL/cmH_2O proposed as poor.
7. Storage pressure > 40 cmH_2O associated with upper-tract damage.
8. DLPP > 40 cmH_2O in myelodysplasia: hydronephrosis/VUR in 85% (McGuire).
9. ALPP < 60 cmH_2O suggests ISD; ≥ 90 suggests hypermobility.
10. VLPP is lower than cough LPP in the same patient.
11. MUCP ≤ 20 cmH_2O defines ISD (type III SUI).
12. Up to 50% of women with UUI show no DO on CMG
13. Single terminal IDC predicts storage-symptom resolution after TURP.
14. BOOI = PdetQmax − 2Qmax; > 40 obstructed.
15. BCI = PdetQmax + 5Qmax; < 100 weak.
16. BOOI cannot be used in women.
17. Blaivas–Groutz nomogram overdiagnoses female obstruction.
18. VUDS (Nitti) female BOO: radiographic narrowing with sustained contraction of any magnitude.
19. Guarding reflex is normal EMG rise during an IDC.
20. DESD requires a suprasacral lesion.
21. DISD and PBNO are diagnosed only by VUDS.
22. Asymptomatic bacteriuria: give prophylaxis before UDS.
23. Prophylaxis of choice: single oral trimethoprim–sulfamethoxazole within 1 h.
24. Impaired compliance: only absolute urodynamic indication to treat BOO.
25. EAU PFS trigger: Qmax > 10 mL/s (not < 10).
26. Bethanechol test positive: rise ≥ 15 cmH_2O (Cannon's law).
27. VUDS is gold standard invasive UDS in NLUTD.

109 Neuromuscular Dysfunction of the Lower Urinary Tract

Reading time ≈ 42 min (slow pace)

1 · PRINCIPLES AND CLASSIFICATION

- Neuropathic lower urinary tract (LUT) dysfunction is largely **predictable from lesion site and nature** (destructive, inflammatory, irritative) — usually, not always. Acute dysfunction may differ from chronic.
- Describe every disorder by **five International Continence Society (ICS) parameters**: sensation (normal/absent/impaired); detrusor (normal, overactive, areflexic, impaired contractility); compliance (normal/↓/↑); smooth sphincter (synergic/dyssynergic); striated sphincter (synergic, dyssynergic, bradykinetic, impaired voluntary control, fixed tone).
- **Neuroplasticity**: structural/neurochemical change reversible up to a point, then fixed. In bladder outlet obstruction (BOO): hypertrophy, ↑ type III:I collagen (↓ compliance), partial denervation, and nerve growth factor (NGF)–driven afferent sensitisation → detrusor overactivity (DO). This is why symptoms persist after urodynamically successful de-obstruction — the lesion starts dysfunction, plasticity maintains it.

Level	Typical disorders	Detrusor	Smooth sphincter	Striated sphincter
Suprapontine	Stroke, tumour, Parkinson disease (PD), dementia, cerebral palsy	Overactive	Synergic	Synergic (± impaired voluntary control; PD bradykinesia)
Multiple system atrophy (MSA)	—	Overactive → impaired	**Open**	Denervated
Suprasacral cord	Spinal cord injury (SCI), multiple sclerosis (MS), transverse myelitis	Overactive	Synergic (dyssynergic if above T7)	**Dyssynergic**
Sacral cord / cauda equina	Sacral SCI, disc, myelodysplasia	Areflexic	Competent non-relaxing; may open	Fixed tone
Peripheral	Radical pelvic surgery, diabetes, tabes	Impaired / areflexic	Open (pelvic surgery)	Fixed tone / synergic

> **KEY POINT** Dyssynergia appears only when the lesion lies **between the pons and the sacral cord**; an open bladder neck marks autonomic (MSA, radical pelvic surgery, myelodysplasia) or sacral damage. PD and MSA share the detrusor pattern but **not the outlet**.

2 · LESIONS AT OR ABOVE THE BRAINSTEM

Cerebrovascular accident (stroke)

- Urinary incontinence (UI) persisting 7 days predicts poor survival; commonest type = impaired-awareness incontinence > urgency UI (UUI).
- **Acute**: retention from areflexia ("cerebral shock") → **chronic**: DO with intact sensation and **synergic sphincters**. If the patient can voluntarily contract the striated sphincter → urgency/frequency only; if not (cortex/internal capsule lesion) → UUI. Basal ganglia/thalamic lesions keep voluntary control.
- **True detrusor–sphincter dyssynergia (DSD) does not occur**: apparent "dyssynergia" is **pseudodyssynergia** — electromyographic (EMG) flare from voluntary attempts to inhibit an involuntary contraction.

- Men with pre-stroke LUT symptoms (LUTS): poor flow/high post-void residual (PVR) usually prostatic, but do full urodynamic study (UDS) before outlet surgery to exclude DO with impaired contractility.

Dementia, brain injury, tumour, others

- **Dementia**: UI from DO or a normal reflex at the wrong time/place (no reason to inhibit). Anticholinergics worsen cognition, delirium, falls and raise new-onset dementia risk → **β3 agonist strongly preferred** (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction [SUFU]).
- **Traumatic brain injury**: above the pontine micturition centre (PMC) → DO, synergic; isolated brainstem injury below PMC → DSD possible.
- **Brain tumour**: site, not histology (superior frontal lobe) — DO, patient unaware of LUT events; frontal lesions can cause isolated retention.
- **Normal-pressure hydrocephalus**: dementia + gait ataxia + UI (DO); ventriculoperitoneal shunt improves UUI.
- **Cerebral palsy**: most have control; LUTS ~half; DO with synergic sphincters; adults with LUTS — poor compliance/detrusor leak point pressure (DLPP) ≥40 cm H_2O in about half → full evaluation. Conservative care succeeds in ~¾.

Parkinson disease vs multiple system atrophy

- **PD**: dopaminergic loss in substantia nigra removes basal-ganglia inhibition of the micturition reflex (loss of inhibitory D1, unopposed facilitatory D2). LUTS 35–70%, storage symptoms dominate. UDS: DO; sphincter **bradykinesia** and sporadic activity mimic DSD but are not obstructive.
- Treatment: overactive bladder (OAB) drugs with low central nervous system (CNS) effect (mirabegron, trospium, darifenacin, extended-release fesoterodine); onabotulinumtoxinA; posterior tibial nerve stimulation (PTNS) and sacral neuromodulation (SNM) effective; subthalamic deep brain stimulation improves capacity.
- **transurethral resection of the prostate (TURP) in PD** (transurethral resection of prostate): not contraindicated — old "post-TURP incontinence in PD" cases were MSA. Confirm PD, exclude MSA, do videourodynamics first.
- **MSA**: glial α-synucleinopathy, cell loss includes intermediolateral columns and **Onuf nucleus** → early severe LUTS + erectile dysfunction (ED); DO early, poor emptying late. Treating DO or sphincteric UI worsens emptying → storage + clean intermittent catheterisation (CIC); desmopressin for nocturia. **Prostatectomy hazardous.**

Feature	MSA	PD
Urinary symptoms / ED	Precede or accompany parkinsonism; ED near universal	Years after diagnosis; ED ~⅓
Dominant problem; PVR	UI; high PVR	Frequency/urgency; PVR normal
Bladder neck	**Open**	Synergic
Striated sphincter EMG	**Denervation** (motor-unit)	Normal (bradykinesia)
Detrusor	DO → areflexia	DO; areflexia uncommon
Outlet surgery	Worsens	Acceptable after work-up

3 · SPINAL CORD DISEASE

Multiple sclerosis

- Demyelination with axonal sparing, almost always cervical cord. LUTS 50–90%; presenting complaint in up to 15%.

- UDS: DO commonest (~⅔); DSD in 30–65% of those with DO; areflexia/impaired contractility in a quarter; sensation usually intact (helps separate pseudodyssynergia).
- Complication risk factors: **DSD in men, high filling pressure, indwelling catheter**. Upper-tract damage less common than SCI; symptoms do not track disease → **baseline UDS essential**.
- Management: anticipatory, **reversible** (pattern changes over time) — antimuscarinic + CIC cornerstone; onabotulinumtoxinA durable; SNM in selected (magnetic resonance imaging [MRI]-compatible devices). Credé/Valsalva contraindicated; indwelling catheter last resort, suprapubic if used.

Spinal cord injury

- State column vs cord level: sacral cord begins at column T12–L1; conus → cauda equina ~L2. Stones: risk highest in first 3 months (immobilisation hypercalciuria).
- **Spinal shock**: depressed cord excitability at/below lesion → areflexic, acontractile smooth bladder; bladder neck closed, some striated activity, no voluntary control → retention, overflow only. **Catheterise** (CIC preferred) keeping pressure low. Lasts usually **6–12 weeks** (up to 1–2 years); reflex contractions return with lower-limb deep tendon reflexes.
- **Suprasacral SCI**: DO (C-fibre–mediated reflex emerges through afferent sprouting, NGF) + smooth synergia + striated DSD → storage *and* emptying failure at high pressure. Lesions at or above cord **T7–T8** (column T6) may add smooth sphincter dyssynergia.
- **Sacral SCI**: areflexia, compliance normal then may fall (decentralisation plasticity); competent non-relaxing bladder neck (may later open); fixed striated tone. Silent upper-tract damage can occur **without vesicoureteral reflux (VUR)**. Straining/Credé meets a fixed outlet.

	Suprasacral (complete)	**Sacral (complete)**
Detrusor	Overactive	Areflexic
Compliance	Normal initially	Normal/high → may fall
Smooth sphincter	Synergic (dyssynergic above T7–T8)	Competent non-relaxing → may open
Striated sphincter	Dyssynergic	Fixed tone, no voluntary control
Somatic signs	Spasticity, hyperreflexia	Flaccidity, ↓ reflexes
Management	Lower pressure (drugs/toxin) + CIC	CIC + keep compliance safe

- **Neurology ≠ urodynamics**: somatic completeness ≠ autonomic completeness; occult second levels. DSD occurs only with suprasacral cord lesions, never suprapontine → coordination is supraspinal (pontine). Thoracolumbar (T10–L2 column) lesions least predictable. **Manage by UDS, not by neurological level**; one study may not suffice.
- Upper-tract risk: detrusor fibrosis, leak point pressure ≥75 cm H_2O, DO for >⅓ of filling → treat neurogenic DO (NDO) aggressively. †DLPP >40 cm H_2O is the classic threshold for upper-tract deterioration.
- Management (suprasacral): if pressures low or can be made low → treat as emptying failure with CIC; antimuscarinic and/or β3 agonist, intradetrusor onabotulinumtoxinA if refractory. Sphincterotomy or intrasphincteric toxin (men) → convert to storage failure with condom collector, only if unable/unwilling to do CIC.

Autonomic dysreflexia (AD)

- Potentially fatal emergency in SCI **above T6** (above splanchnic sympathetic outflow); after spinal shock, may appear years later. Noxious stimulus below lesion → unopposed spinal sympathetic reflex (no descending medullary inhibition).
- Features: pounding headache, hypertension, flushing and sweating above lesion, **bradycardia**; may cause cerebral haemorrhage, seizure. Triggers: bladder/rectal distension, blocked catheter, instrumentation, stones, sores, fracture, sex.
- UDS risk: complete, cervical, DSD, poor compliance; defined during UDS as systolic rise ≥20 mm Hg → monitor blood pressure (BP) in all SCI studies.

ALGORITHM
†Sit upright, loosen clothing → **remove stimulus** (drain bladder, unblock catheter, rectum) → if BP persists: nitroglycerin 2% paste above lesion (first line) → labetalol / intravenous nitroprusside → prophylaxis: terazosin or prazosin (α1) → intractable: sacral rhizotomy, sympathectomy, cordectomy

- Nifedipine (10–20 mg sublingual during cystoscopy) now banned in many centres — uncontrolled hypotension.

VUR and women with SCI

- VUR in 17–25% of SCI, driven by high pressure and infection; **chronic urethral catheter worst** for upper tracts. Treat the bladder first (lower storage pressure/outlet resistance); anti-reflux surgery in a thick bladder is hard. VUR artefactually enlarges capacity and lowers pressures on UDS.
- Destroyed urethra: bladder neck closure + suprapubic catheter, or diversion.
- Women: no external collecting device — paraplegics master CIC, tetraplegics often need indwelling catheters.

Other cord disorders

- **Acute transverse myelitis**: retention almost universal → DO with DSD; manage as SCI; UDS before irreversible therapy.
- **Cervical myelopathy/stenosis**: DO ± DSD, or underactivity (posterior funiculus) → UDS guides therapy.
- **Neurospinal dysraphism**: myelomeningocele ~90% of spina bifida cystica; level correlates poorly with UDS. Typical adult: **areflexic bladder with open bladder neck**, fixed sphincter → leaks at fixed pressure; DSD in 10–15%. Poor compliance, high pressures and DO predict renal decline. Stress UI: bulking safer than sling/artificial urinary sphincter (does not raise closure pressure enough to reduce compliance). Medical + onabotulinumtoxinA → augmentation/diversion.
- **Tethered cord**: presents after growth spurts; no typical pattern (DO commonest); urinary recovery after release lags pain and motor.
- **Sensory neurogenic bladder** (tabes, B12 deficiency/subacute combined degeneration): lost sensation → large capacity, decompensated detrusor; B12 replacement may reverse. **Motor neurogenic bladder** (poliomyelitis): retention with intact sensation.

4 · LESIONS DISTAL TO THE CORD

- **Disc disease**: central prolapse (L4–L5, L5–S1) compresses cauda equina. Commonest finding: **normally compliant areflexic bladder**; patients strain or present in retention. Compliance preserved (unlike myelomeningocele) because disc injury is incomplete for preganglionic parasympathetic motor fibres. **Pre-laminectomy UDS** (clinical and medicolegal).
- **Cauda equina syndrome**: saddle anaesthesia, loss of voluntary anal/urethral sphincter control and sexual response; acontractile detrusor. **Decompression within 48 h** gives better urinary and rectal recovery. Sphincter recovers before detrusor.
- **Radical pelvic surgery** (abdominoperineal resection [APR], radical hysterectomy): pelvic plexus injury. Permanent pattern: impaired contractility + fixed striated tone + **open non-functional bladder neck** + ↓ compliance → leak with abdominal pressure or retention; positive bethanechol supersensitivity. Early: **CIC and wait 6–12 months**. **Avoid prostatectomy unless BOO proven**. Nerve-sparing radical hysterectomy reduces long-term urinary symptoms.
- **Herpes zoster** (sacral dorsal root ganglia): retention days–weeks after rash; resolves in 1–2 months. **Herpes simplex**: retention + cerebrospinal fluid (CSF) pleocytosis + sacral radiculitis = **Elsberg syndrome**, transient.
- **Diabetic cystopathy**: sensory then motor neuropathy → ↓ sensation → ↑ capacity → ↓ contractility → ↑ PVR. In men, pressure–flow studies separate it from BOO; "dyssynergia" is straining artefact.

DO also common (early phase/coexisting cerebrovascular disease) → UDS before treatment. Timed voiding prevents distension decompensation.

- **Guillain–Barré syndrome**: ↑ PVR, retention; manage reversibly (CIC) pending recovery.

> **KEY POINT** Distal lesions hit the peripheral reflex arc → areflexia with a non-relaxing/fixed outlet; compliance falls only when preganglionic parasympathetic motor fibres are damaged. Hence urgent decompression, waiting and CIC outrank outlet surgery.

5 · SPECIFIC FUNCTIONAL AND MISCELLANEOUS CONDITIONS

Detrusor–sphincter dyssynergia

- True DSD requires a lesion **between sacral cord and PMC** (SCI, MS, transverse myelitis); without one, doubt the diagnosis (exception: Hinman syndrome in children).
- Pseudodyssynergia: abdominal straining or attempted inhibition of a contraction — misdiagnosis leads to harmful treatment.
- Untreated, ~half of men develop VUR, upper-tract damage, stones, urosepsis; far less in women. DLPP ≥40 cm H_2O after sphincterotomy = failure.
- Treatment: oral sphincter drugs disappoint. **CIC + DO control preferred**; sphincterotomy or sphincteric toxin (poor durability); indwelling catheter; diversion. Urethral stents abandoned.

Blaivas type	EMG pattern	Lesion	Treatment
1	Activity rises with detrusor pressure, relaxes suddenly at peak	Incomplete	Observe unless VUR, hydronephrosis, AD
2	Sporadic contractions throughout	Complete	Treat
3	Crescendo–decrescendo, obstruction throughout	Complete	Treat

Functional outlet disorders

- **Dysfunctional voiding** (Hinman, "non-neurogenic neurogenic bladder"): involuntary striated sphincter contraction during voiding without neurological disease, proven by pressure–flow–EMG without straining.
- **Non-neurogenic chronic retention**: PVR ≥**300 mL** persisting ≥6 months on ≥2 occasions. High risk (hydronephrosis, chronic kidney disease, recurrent urinary tract infection [UTI], UI with skin sequelae) → drainage; suprapubic preferred to urethral long term.
- **Primary bladder neck obstruction**: young/middle-aged men, long-standing symptoms, normal prostate and cystoscopy. Diagnose by videourodynamics (BOO localised to bladder neck). α-blockers help some; **bladder neck incision definitive**. "Trapped prostate" (Turner-Warwick) → small TURP including bladder neck.
- **Female BOO**: rare; Groutz criteria — free maximum flow (Qmax) ≤12 mL/s + detrusor pressure at Qmax ≥20 cm H_2O. Bladder neck surgery risks sphincteric UI.
- **Fowler syndrome**: retention in young women (capacity >1 L without urge), normal MRI, often polycystic ovaries; concentric needle EMG — **complex repetitive discharges and decelerating bursts**; urethral closure pressure >100 cm H_2O. Hormones, drugs and toxin fail; **SNM is the treatment** (~70%), and Fowler syndrome predicts SNM response.

Other conditions — key points

- **Myasthenia gravis**: high incontinence after prostatectomy. **Isaacs syndrome**: sphincter spasm (catheter will not pass) → suprapubic drainage, plasmapheresis. **Wernicke**: DO or poor contractility, resolve with thiamine.
- Myelopathies (tropical spastic paraparesis, schistosomal, syringomyelia) → DO + DSD; root disease (cytomegalovirus, Lyme) → areflexia. **Lupus cystitis**: thick wall, hydronephrosis → steroids.
- **Radiation**: early storage symptoms recover by 6 months; late effects progressive.
- **Defunctionalised bladder**: rehabilitate by cycling; transplant feasible without augmentation.

6 · EVALUATION AND RISK STRATIFICATION

- Guidelines: American Urological Association (AUA)/SUFU neurogenic LUT dysfunction (NLUTD) 2021; European Association of Urology (EAU) Neuro-urology 2022. **Primary goal: prevent renal damage.**
- All: history (neurological condition, bladder management, goals), examination (cognition, mobility, **hand dexterity**, spasticity, sensation, skin), urinalysis; PVR if voiding; diaries.
- **Defer full assessment during spinal shock or acute brain injury**; no irreversible therapy in this phase.

Risk (AUA/SUFU)	Typical	Defining findings
Low (all)	Suprapontine or uncomplicated peripheral	Spontaneous voiding, low PVR, stable; normal imaging/renal function
Unknown	Infrapontine–suprasacral; sacral/infrasacral	Storage + voiding symptoms, ↑ PVR, recurrent UTI; not yet assessed → UDS, upper-tract imaging, renal function
Moderate	—	Neurogenic retention; DO with incomplete emptying; normal tracts
High (any one)	—	Poor compliance, ↑ storage pressure, DO with DSD, VUR; hydroureteronephrosis, scarring, large stones; abnormal renal function

KEY POINT Low risk needs no routine imaging, renal function or UDS. DO alone fits low risk (suprapontine, synergic); DO in the high-risk row means DO with infrapontine pressure consequences. Risk is judged by **pressure, compliance and upper tracts**, not by diagnosis.

7 · MANAGEMENT

- **Goals**: protect upper tracts; control infection; low-pressure storage and emptying; continence and control; avoid catheter/stoma if possible; social acceptability.
- **Principles**: start with simplest, most reversible therapy and climb; combine mechanisms; shared decisions (dexterity, cognition, carers, prognosis).

ALGORITHM
Low-pressure storage + regular emptying: CIC + antimuscarinic / β3 agonist → intradetrusor onabotulinumtoxinA → neuromodulation (selected) → augmentation cystoplasty / continent diversion → bladder neck closure + suprapubic catheter / ileovesicostomy / incontinent diversion (last resort)

Emptying

- **CIC is the gold standard**; keep volumes <**500 mL**. Credé/Valsalva not recommended (outlet resistance → high pressure).
- CIC vs indwelling urethral catheter: less pyelonephritis, epididymo-orchitis, urosepsis. If chronic catheter needed: **suprapubic > urethral** (avoids urethral erosion).

Storage

- Antimuscarinics first-line for NDO (mirabegron effective); refractory → **intradetrusor onabotulinumtoxinA** (~60% continue long term); warn about CIC.
- DO + DSD: detrusor relaxation + CIC preferred; sphincterotomy only for selected men (high failure, repeats); intrasphincteric toxin not routine.
- Neuromodulation: **sacral deafferentation + anterior root stimulator (Brindley)** in complete SCI — continence, fewer UTIs, stable kidneys, AD resolves; rhizotomy benefits even if stimulator off → *Ch 115*. PTNS for voiding patients with OAB (MS, PD, stroke); SNM for NLUTD without SCI or spina bifida.

- Reconstruction (failed less invasive options, e.g. cannot catheterise): bladder neck closure (retropubic better than perineal) + suprapubic catheter; augmentation + fascial sling; ileovesicostomy.

Urinary tract infection

- **Treat bacteriuria only if symptomatic.** Correct risk factors (stones, residuals). **No prophylaxis** for CIC or indwelling catheter users. Refractory → upper-tract imaging, cystoscopy, UDS.

Bladder cancer

- Historically squamous carcinoma with long-term catheters; contemporary data: incidence comparable to general population but often muscle-invasive at diagnosis; catheters and UTI, not SCI, are the risk. **AUA/SUFU 2021: no routine surveillance cystoscopy**; scope for haematuria.

8 · FOLLOW-UP

Body	Schedule
AUA/SUFU 2021 (risk-based)	Low risk: none routine. Moderate/high: yearly history, examination, renal function; upper-tract imaging every 1–2 years; repeat UDS for change; bowel reconstruction: yearly electrolytes/acid–base, imaging
EAU	Specialist review every 1–2 years; ultrasound + residual 6-monthly; dipstick 2-monthly; yearly examination and bloods
American Paraplegia Society (SCI)	Annual for 5–10 years then alternate years; renogram — >20% fall in renal plasma flow → investigate; annual cystoscopy with indwelling catheter (**not endorsed** by AUA/SUFU)

9 · MCQ ONE-LINERS

1. Suprapontine lesion: DO with synergic sphincters; true DSD never occurs.
2. DSD requires a lesion between the PMC and the sacral cord.
3. Smooth sphincter dyssynergia added in SCI at or above cord T7–T8 (column T6).
4. Open bladder neck + denervated striated sphincter = MSA, not PD.
5. Post-TURP incontinence in "PD" historically reflected undiagnosed MSA.
6. Preferred OAB drug in dementia: β3 agonist.
7. Spinal shock usually lasts 6–12 weeks; bladder areflexic, bladder neck closed.
8. Autonomic dysreflexia: SCI above T6; hypertension with bradycardia.
9. First-line drug for acute AD: nitroglycerin 2% paste after removing stimulus.
0. Blaivas type 1 DSD: incomplete lesion, sphincter relaxes at peak detrusor pressure.
1. Untreated DSD: about half of men develop upper-tract complications.
2. Commonest UDS finding in lumbar disc disease: normally compliant areflexic bladder.
13. Cauda equina: decompression within 48 h improves urinary recovery.
14. Radical pelvic surgery: open bladder neck, fixed striated tone, ↓ compliance.
15. Elsberg syndrome: herpes simplex sacral radiculitis with retention and CSF pleocytosis.
16. Classic diabetic cystopathy: ↓ sensation, ↑ capacity, ↓ contractility, ↑ PVR.
17. Typical adult myelodysplastic bladder: areflexic with open bladder neck.
18. Fowler syndrome EMG: complex repetitive discharges and decelerating bursts; treat with SNM.
19. Non-neurogenic chronic retention: PVR ≥300 mL for ≥6 months on ≥2 occasions.
20. CIC volumes should stay below 500 mL.
21. No antibiotic prophylaxis for CIC or indwelling catheter users.
22. AUA/SUFU 2021: no surveillance cystoscopy in NLUTD.
23. Sacral deafferentation with Brindley stimulator abolishes AD and NDO.
24. Primary bladder neck obstruction: definitive treatment is bladder neck incision.

113 Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure

Reading time ≈ 19 min (slow pace)

1 · FRAMEWORK

Drugs are filed by the Wein functional scheme → *Ch 104*: **storage failure** — relax the bladder/↑ capacity or ↑ outlet resistance; **emptying failure** — ↑ detrusor contraction (weak tools) or ↓ outlet resistance. Most lower urinary tract (LUT) drug failures are of **selectivity/therapeutic window** — the same receptor in heart and vessels caps the dose.

2 · ANTIMUSCARINICS (OVERACTIVE BLADDER, OAB; DETRUSOR OVERACTIVITY, DO)

† Detrusor contraction is mediated by **M3** receptors although **M2** predominate numerically. Antimuscarinics act mainly in the **storage phase** (↓ urgency, ↑ capacity — partly by damping afferent signalling), not by blocking voiding contraction, so retention is uncommon at clinical doses.

Agent †	Usual dose	Point to remember
Oxybutynin	5 mg 2–3×/day; extended release (ER); transdermal	Highest dry mouth (active metabolite N-desethyloxybutynin); transdermal bypasses first-pass → less dry mouth; most central nervous system (CNS) penetration — avoid in elderly
Tolterodine	2 mg bd; ER 4 mg od	Non-selective; bladder > salivary in vivo
Fesoterodine	4–8 mg od	Prodrug of tolterodine's active metabolite; flexible dose
Solifenacin	5–10 mg od	Moderately M3-selective; long half-life
Darifenacin	7.5–15 mg od	Most M3-selective; constipation
Trospium	20 mg bd	**Quaternary amine** — minimal blood–brain barrier crossing; renally excreted
Propiverine, imidafenacin	20 mg; 0.1–0.2 mg bd	Mixed action / Japan

- † **Side effects by receptor**: dry mouth (M3 salivary), constipation (M3 gut), blurred vision (M3 ciliary), tachycardia (M2), cognitive impairment (**M1**, brain).
- † **Contraindications**: **uncontrolled narrow-angle glaucoma**, urinary retention, gastric retention, myasthenia gravis, severe ulcerative colitis/toxic megacolon.
- † **Elderly**: cumulative **anticholinergic burden** is associated with cognitive decline and dementia → prefer β3-agonist or non-CNS-penetrant agent; avoid oxybutynin (Beers criteria).
- † Men with bladder outlet obstruction (BOO): safe if post-void residual (PVR) is low; check PVR before and during treatment.

ALGORITHM
† OAB: behavioural therapy (bladder training, fluids, pelvic floor) → antimuscarinic *or* β3-agonist (β3 first in elderly/cognitive risk) → switch or combine → onabotA 100 U / sacral neuromodulation / percutaneous tibial nerve stimulation → augmentation cystoplasty

3 · B3-ADRENOCEPTOR AGONISTS

† β3 stimulation relaxes the detrusor during storage (↑ cyclic adenosine monophosphate) without the antimuscarinic side-effect profile — no dry mouth, no cognitive burden.

Agent †	Dose	Cautions
Mirabegron	25 → 50 mg od	Avoid in **severe uncontrolled hypertension** (≥**180/110** mm Hg); monitor blood pressure; moderate CYP2D6 inhibitor (metoprolol, desipramine); monitor digoxin

Agent †	Dose	Cautions
Vibegron	75 mg od	No titration; no clinically relevant CYP2D6 inhibition or blood-pressure effect in trials

4 · COMBINATION THERAPY

Different mechanisms justify an antimuscarinic + β3-agonist, especially when antimuscarinic dose gives intolerable adverse events (AEs) or insufficient response.

Direction	Evidence	Agents
Mirabegron added to failing antimuscarinic	SYNERGY, SYNERGY II, BESIDE, MILAI	**Solifenacin only**
Antimuscarinic added to failing mirabegron	MILAI II (52 weeks)	Solifenacin 5, propiverine 20, imidafenacin 0.2, tolterodine ER 4 mg
Vibegron + antimuscarinic	Phase IIb (vibegron 100 + tolterodine 4 mg)	Benefit extrapolated; no trial after monotherapy failure

KEY POINT Combination beats solifenacin 5 mg and either monotherapy; in non-responders it **equals** solifenacin 10 mg but with less dry mouth — it gains **tolerability, not efficacy**, over dose escalation. No change in heart rate, blood pressure, QT or PVR; constipation roughly doubles. Benefit irrespective of age ≥65 or ≥75.

- **α-blocker + desmopressin**: for nocturia resistant to first-line therapy — ≈30% of men ≥60 with benign prostatic hyperplasia (BPH) have nocturnal polyuria, which α-blockade does not treat. Evidence uncontrolled.
- **Vaginal oestrogen + antimuscarinic**: positive signals weak and sequence-dependent (antimuscarinic after oestrogen); simultaneous-combination randomised controlled trials (RCTs) null; vaginal estradiol ≈ oxybutynin. Additive benefit **not established**.
- † Systemic oestrogen worsens incontinence — do not use for urinary incontinence (UI).

5 · INTRADETRUSOR ONABOTULINUMTOXINA (ONABOTA)

† Blocks presynaptic acetylcholine release (cleaves SNAP-25) and reduces afferent signalling; cystoscopic injection into the detrusor.

	Idiopathic OAB (refractory)	Neurogenic DO (NDO; spinal cord injury, SCI; multiple sclerosis, MS)
Dose	**100 U** † (≈20 sites)	**200 U** † (≈30 sites)
Trial	EMBARK — anticholinergic-refractory only	DIGNITY — equal efficacy in anticholinergic users and non-users
Retention	† Raised PVR/need for clean intermittent catheterisation (CIC) — higher in NDO non-catheterisers (MS); patient must accept CIC before treatment	

- † Commonest AE: urinary tract infection (UTI) — treat active UTI first; check PVR ≈2 weeks after. Effect lasts ≈6–9 months; repeat when it wanes (≥12 weeks apart).
- † Caution: myasthenia gravis, aminoglycosides.
- Adding oral drugs to prolong onabotA effect is common but unevaluated.

6 · NOCTURIA — DESMOPRESSIN †

- Vasopressin V2 agonist → ↓ nocturnal urine production; indicated for **nocturnal polyuria**. Low-dose sublingual formulations: women 25 µg, men 50 µg.
- **Hyponatraemia** is the key risk, rising with age **≥65**: check sodium at baseline, within 1 week and at 1 month; stop if low.
- Contraindications: hyponatraemia, polydipsia, heart failure, syndrome of inappropriate antidiuretic hormone, renal impairment, concurrent loop diuretics.

7 · INCREASING OUTLET RESISTANCE (STRESS URINARY INCONTINENCE, SUI)

† Pelvic floor muscle training and surgery remain first-line for SUI; no drug is a standard of care. α-agonists fail because systemic agents lack urethral selectivity — vascular α-receptors raise blood pressure before urethral closure improves.

Drug	Mechanism	Status
Duloxetine †	Serotonin–noradrenaline reuptake inhibitor at **Onuf's nucleus** → ↑ pudendal tone of striated sphincter	40 mg bd; nausea → frequent discontinuation; approved in Europe, not USA
Phenylpropanolamine, midodrine, norepinephrine	Systemic α-agonists	Weak evidence; hypertension, haemorrhagic stroke → phenylpropanolamine banned by US Food and Drug Administration (FDA)
Imipramine †	Tricyclic: anticholinergic + α-agonist-like outlet effect	Mixed UI, enuresis; orthostatic hypotension, cardiac arrhythmia — caution in elderly
Clenbuterol	β-agonist	Small positive RCT, unbalanced baseline; tremor; abandoned
PSD503 vaginal phenylephrine gel	Local α-agonist	↓ leakage but poor acceptability; abandoned

8 · FACILITATING EMPTYING

- † **Bethanechol** (muscarinic agonist): **no convincing controlled evidence** of improved emptying in detrusor underactivity; cholinergic AEs (sweating, bradycardia, bronchospasm, diarrhoea). Contraindicated in asthma, peptic ulcer, bradycardia, mechanical obstruction.
- † **α-blockers** (tamsulosin, alfuzosin, silodosin, doxazosin, terazosin): ↓ smooth-sphincter/prostatic tone — BPH, primary bladder neck obstruction, neurogenic LUT dysfunction; aid trial without catheter after acute retention.
- † Striated sphincter: baclofen, benzodiazepines, dantrolene — limited efficacy; sphincteric onabotA.

Drugs that cause retention †

Anticholinergics (incl. antihistamines, tricyclics, antipsychotics, antiparkinsonian drugs); α-agonist decongestants (pseudoephedrine); opioids; benzodiazepines; anaesthesia/epidurals. Ask about them in every retention.

9 · FAILED DRUGS

Agent	Indication	Reason for failure
Vitamin D3, elocalcitol	UUI/OAB	↓ incontinence only; no urodynamic effect
Calcium antagonists (verapamil, nimodipine)	OAB	No effect; cardiovascular users show no bladder impairment
K^+ channel openers (ZD0947, flupirtine)	OAB	Dose capped by heart; flupirtine hepatotoxic
URO-902 (big potassium [BK] channel gene therapy)	OAB	Phase 1 only
RimabotulinumtoxinB	Toxin A-refractory DO	Short duration (≤3 months)
Intraprostatic onabotA, PRX302, NX-1207	BPH-related lower urinary tract symptoms (LUTS)	= sham injection — large placebo effect

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SYNERGY / II	Phase III, 12 weeks / 12 months	Solifenacin 5 + mirabegron 50 > monotherapy for UI	Combination effective and safe long-term
BESIDE	Solifenacin 5 non-responders	Add-on > solifenacin 5, = solifenacin 10	Alternative to dose escalation
MILAI II	52 weeks, mirabegron non-responders	Four antimuscarinic add-ons safe, effective	Add antimuscarinic to mirabegron
DIGNITY	Pooled RCTs, NDO	OnabotA 200 U equal in anticholinergic users/non-users	Efficacy independent of anticholinergic
EMBARK	RCTs, refractory OAB	OnabotA 100 U effective	Standard dose for idiopathic OAB

11 · MCQ ONE-LINERS

1. Receptor mediating detrusor contraction: M3.
2. Most M3-selective antimuscarinic: darifenacin.
3. Quaternary amine, minimal CNS penetration: trospium.
4. Transdermal oxybutynin: less dry mouth — avoids N-desethyloxybutynin.
5. Fesoterodine shares tolterodine's active metabolite.
6. Absolute antimuscarinic contraindication: uncontrolled narrow-angle glaucoma.
7. Cognitive side effects: M1 blockade.
8. Mirabegron caution: severe uncontrolled hypertension.
9. Mirabegron inhibits CYP2D6.
10. Mirabegron add-on evidence exists only with solifenacin.
11. Combination vs solifenacin 10 mg: equal efficacy, less dry mouth.
12. OnabotA dose: 100 U idiopathic OAB, 200 U NDO.
13. Commonest onabotA AE: UTI.
14. Desmopressin key risk: hyponatraemia, especially age ≥65.
15. Duloxetine acts at Onuf's nucleus.
16. Commonest duloxetine AE: nausea.
17. Phenylpropanolamine banned: haemorrhagic stroke.
18. Bethanechol: no convincing evidence for detrusor underactivity.
19. Imipramine: anticholinergic plus outlet α-effect.
20. Intraprostatic toxins failed against sham injection.
21. RimabotulinumtoxinB limitation: short duration.
22. Flupirtine halted for hepatotoxicity.

115 Electrical Stimulation and Neuromodulation in Storage and Emptying Failure

Reading time ≈ 18 min (slow pace)

1 · EVOLUTION AND MECHANISM

- Direct detrusor/sphincter stimulation failed — painful, only locally effective, and bladder and sphincter contracted together. **Indirect stimulation via afferents** needs less current and treats the whole organ → "neuromodulation".
- Milestones [MCQ]: sacral neuromodulation (SNM) ~1985 (Tanagho, Schmidt); first tibial nerve stimulation 1983 (McGuire); percutaneous tibial nerve stimulation (PTNS) derived from acupuncture point **SP6**; first tibial implant 2006.
- **Exclude anatomical causes first** (stones, cancer, fistula, poorly compliant bladder, stricture, prostatic enlargement, prolapse): stimulation acts only on neural control, never on anatomy or muscle bulk.
- Normal voiding: external urethral sphincter (EUS) relaxes, detrusor contracts 1–2 s later; the pontine micturition centre (PMC) acts as an on-switch, gated by EUS relaxation.
- **Current theory (forebrain)**: sacral roots are mixed; stimulation excites afferents → forebrain areas of awareness and alertness → less PMC firing in overactive bladder (OAB). The older efferent theory is less plausible: benefit needs no pelvic floor contraction, and purely sensory nerves work equally.
- **Paradox**: the same therapy treats underactive bladder — raised awareness of filling lets the PMC switch on; in retention SNM works mainly by **striated sphincter relaxation**, not detrusor contraction.

2 · SACRAL NEUROMODULATION

Indications

- **Third-line** for idiopathic OAB after lifestyle/pelvic floor training and drugs (American Urological Association [AUA]/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction [SUFU] and European Association of Urology [EAU] identical).
- Pre-treatment: frequency–volume diary (OAB definition ignores excess intake); exclude urinary tract infection (UTI), significant post-void residual (PVR), stones, cancer. Urodynamic detrusor overactivity (DO) **does not predict** success.
- **Non-obstructive urinary retention (NOUR)**: chronically high PVR needing drainage; PVR is the only objective outcome. Fowler syndrome predicts response; increasing age lowers odds.
- †Faecal incontinence is an established, approved SNM indication.

Food and Drug Administration (FDA) approved	Not approved
Urgency urinary incontinence (UUI)	Bladder pain syndrome (offer selectively if coexisting OAB)
Urgency–frequency	Pelvic pain syndrome
Chronic NOUR	Neurogenic bladder (but pooled success comparable)

Test phase — selection decides success

- Every patient has a test. **Success = ≥50% improvement** in ≥1 diary parameter (frequency, voided volume, leaks) vs a 3–5-day baseline; retention — catheterisation reduced ≥50%, ideally none.
- **Percutaneous nerve evaluation (PNE)**: local anaesthesia → needle into **S3** by landmarks (≈9 cm above coccyx tip, 1–2 cm lateral; fluoroscopic cross-hair optional) → sensory then motor responses → temporary lead, external stimulator, 5–7 days. Positive → full implant in one session.

- **Staged implant**: stage 1 — permanent **tined quadripolar lead** into S3 under fluoroscopy, tunnelled extension, test up to 4 weeks → stage 2 at 2–4 weeks: generator in buttock. Failure → remove lead. Aim: responses on all four contacts at low amplitude. Prophylaxis: cefazolin + metronidazole (± meticillin-resistant *Staphylococcus aureus* cover).

	PNE	**Staged**
Setting	Office, no incision	Theatre
Lead	Temporary, migrates → false-negative test	Permanent tined, precise
Test	5–7 days	Up to 4 weeks
Conversion	Lower	Higher
NOUR	Not used	Used

Root	**Perineal/anal**	**Leg/foot**
S2	—	Calf/foot external rotation
S3 (target)	Anal "bellows", perineal/genital sensation	**Great-toe flexion**
S4	Perineal/anal response	None

Device and programming

- Pulse width 210 µs, 10–16 Hz, amplitude at sensory threshold. Reprogramming ≈3 times in year 1. Cycling stimulation equals continuous. **Bilateral not superior** to unilateral (randomised controlled trial [RCT]).
- Battery: non-rechargeable ≥**10 years**, rechargeable ≥15 years.
- **Magnetic resonance imaging (MRI)**: current devices conditional for whole-body MRI at **1.5 and 3 T**; at replacement remove all old components.

Complications

- Revision ~3%, explantation ~9% at 2 years (ROSETTA); loss of efficacy; lead dislocation; infection.
- †Implant-site pain, unwanted stimulation/leg pain — managed by reprogramming, then revision; infection → explant.

3 · TIBIAL NERVE STIMULATION

- Posterior tibial nerve carries S2–S3 fibres (shared with bladder). Mechanism: cortical reorganisation; in animal OAB models inhibition is opioid-mediated (C fibres; blocked by naloxone).

ALGORITHM
Supine → 34-gauge needle ≈3–4 cm (three fingerbreadths) above medial malleolus, behind tibia, at 60–90° → ground pad same leg → 20 Hz, 200 µs, 0–10 mA → confirm: **great-toe flexion/toe fanning** (motor) or sole tingling (sensory) → **12 weekly sessions × 30 min** → taper → maintenance

- Indications: OAB refractory to or intolerant of drugs (office-based, no implant); safe in neurogenic OAB (multiple sclerosis, Parkinson disease, stroke); children failing bladder training and drugs.
- Success: ≥50% fall in urgency/UUI. **Relapse is likely without maintenance**; tapering sustains results to 3 years.
- Vs tolterodine: equal efficacy, fewer side effects. Better outcome without DO; poor mental-health score predicts failure.
- **Transcutaneous (TTNS)**: home surface electrodes; skin impedance limits amplitude → less effective than PTNS, cheaper.
- **Implantable tibial stimulators** (BlueWind, eCoin) remove the clinic burden; PTNS response does not guarantee implant response.

4 · OTHER TARGETS

- **Pudendal nerve** (S2–S4, Alcock canal): tined lead, posterior approach (perineal has more infection). Blinded crossover vs SNM: greater improvement, preferred by most; salvage after failed SNM. Not FDA approved.
- **Dorsal genital nerve**: purely sensory; acutely suppresses neurogenic DO in spinal cord injury (SCI) — suits **conditional** (on-demand) stimulation; limited by electrode fixation and site. Experimental.
- **Sacral dermatome (transcutaneous)**: raises capacity; cannot abort contractions acutely.
- **Intravaginal stimulation** for stress incontinence: safe, evidence of low quality.
- †**Intravesical stimulation**: catheter electrode in saline-filled bladder activates mechanoreceptor afferents to improve sensation and reflex contraction in neurogenic underactive bladder (incomplete lesions; afferent pathway must be intact); evidence limited.

5 · SACRAL ANTERIOR ROOT STIMULATION (BRINDLEY)

- Stimulates **anterior sacral roots** → bladder contraction (voiding), defecation, erection — in complete SCI.
- Stimulation cannot reduce DO → combined with **posterior (dorsal) sacral rhizotomy** of the same roots: abolishes reflex incontinence, raises capacity and compliance, protects kidneys, abolishes autonomic dysreflexia.
- Voiding works because the striated sphincter relaxes faster than the detrusor after each burst → †"post-stimulus voiding". †Cost of rhizotomy: loss of reflex erection, reflex ejaculation and residual sacral sensation.

6 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SUmiT	RCT, PTNS vs validated sham, 12 weeks, n = 220	Responders 55% vs 21%	First sham-proven PTNS efficacy; maintenance keeps 77% relapse-free at 3 years
InSite	RCT, SNM vs standard medical therapy, mild–moderate OAB	Success 61% vs 42% at 6 months	SNM beats continued drugs
ROSETTA	RCT, onabotulinumtoxinA 200 U vs SNM, refractory UUI women	Toxin slightly fewer leaks at 6 months; equal at 2 years; toxin more UTI, some clean intermittent catheterisation	Both valid third-line; registered OAB toxin dose is 100 U

7 · MCQ ONE-LINERS

1. SNM target root: S3 — bellows and great-toe flexion.
2. S2 response: calf/foot external rotation; S4: perineal response, no toe.
3. SNM test success: ≥50% improvement in a diary parameter.
4. Urodynamic DO does not predict SNM success.
5. FDA-approved SNM: UUI, urgency–frequency, NOUR.
6. Neurogenic bladder and pelvic pain: not FDA-approved for SNM.
7. Staged implant preferred for NOUR; PNE not used.
8. Bilateral SNM not superior to unilateral.
9. Modern SNM devices: conditional MRI at 1.5 and 3 T.
10. SNM in retention acts mainly by sphincter relaxation.
11. Fowler syndrome predicts good SNM response.
12. Current SNM theory: afferent modulation of forebrain awareness centres.
13. PTNS: 34-gauge needle, 3–4 cm above medial malleolus, 20 Hz.
14. PTNS protocol: 12 weekly 30-minute sessions, then maintenance.
15. PTNS derived from acupuncture point SP6.

6. First tibial nerve stimulation for bladder: McGuire, 1983.
7. SUmiT: first validated sham; 55% vs 21%.
8. PTNS inhibition of OAB-type reflex is naloxone-sensitive.
9. TTNS less effective than PTNS due to skin impedance.
20. Pudendal stimulation: salvage after failed SNM.
21. Brindley stimulator combined with posterior sacral rhizotomy.
22. ROSETTA used 200 U toxin — twice the registered OAB dose.

★ Putting it together: voiding dysfunction decisions

Reading time ≈ 6 min (slow pace) · Integrates Ch 103–115 · use these as the skeleton of long-essay answers

A · THE FUNCTIONAL CLASSIFICATION AS A TREATMENT MAP

	Because of the bladder	**Because of the outlet**
Failure to store	Overactivity or low compliance → antimuscarinic, β3-agonist, onabotulinumtoxinA, neuromodulation, augmentation	Sphincter weakness → pelvic floor training, duloxetine (Europe), sling, artificial urinary sphincter, bulking
Failure to empty	Underactive or acontractile detrusor → clean intermittent catheterisation (CIC); sacral neuromodulation for non-obstructive retention	Obstruction (anatomical, or dyssynergia) → α-blocker, prostate/bladder-neck surgery, sphincterotomy, CIC

If neither can be fixed, circumvent: CIC, indwelling catheter or urinary diversion (→ *Ch 104*).

B · WHERE IS THE LESION? EXPECTED URODYNAMICS

Lesion	**Detrusor**	**Striated sphincter**	**Key risk**
Above the pons (stroke, Parkinson disease, dementia)	Overactive	Synergic	Urgency incontinence; low upper-tract risk
Spinal cord between pons and sacral cord (spinal cord injury, multiple sclerosis)	Overactive	**Dyssynergic**	High pressure → upper-tract damage; autonomic dysreflexia if lesion above T6
Sacral cord / cauda equina / peripheral (diabetes, pelvic surgery)	Areflexic, often reduced sensation	Fixed tone, cannot relax voluntarily	Retention, overflow; may lose compliance

KEY POINT Dyssynergia appears only when the lesion lies between the pons and the sacral cord, because the pons coordinates detrusor contraction with sphincter relaxation. A detrusor leak point or storage pressure ≥40 cmH_2O, or poor compliance, threatens the kidneys and is the main reason to treat (→ *Ch 107, 109*).

C · OVERACTIVE BLADDER: TREATMENT LADDER

ALGORITHM

1. Behavioural: fluid and caffeine adjustment, bladder training, pelvic floor muscle training; treat constipation; review drugs.

2. Drugs: antimuscarinic or β3-agonist (prefer β3-agonist or a low-CNS antimuscarinic in the elderly); combine if partial response; check post-void residual in men with obstruction (→ *Ch 113*).

3. Refractory: intradetrusor onabotulinumtoxinA (100 U idiopathic, 200 U neurogenic; warn about CIC), percutaneous tibial nerve stimulation, or sacral neuromodulation (staged test; success = ≥50% improvement) (→ *Ch 115*).

4. Last resort: augmentation cystoplasty or urinary diversion.

D · SPINAL CORD INJURY: MANAGING THE BLADDER

ALGORITHM

Spinal shock (areflexic bladder): indwelling or intermittent catheter until reflexes return.

Then urodynamics; CIC + antimuscarinic for a high-pressure overactive bladder; onabotulinumtoxinA if needed; sphincterotomy or diversion if CIC is impossible; lifelong upper-tract surveillance.

Autonomic dysreflexia (lesion above T6): sudden hypertension, bradycardia, headache, flushing above the lesion → sit upright, loosen clothing, find and remove the stimulus (blocked catheter, full bladder, constipation), short-acting antihypertensive if pressure stays high (→ *Ch 109*).